

ESSENTIAL MATHEMATICS GRADE 10: TRIGONOMETRY

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.3 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Essential Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.3: Trigonometry
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Identification of sides of a right-angled triangle (hypotenuse, opposite, adjacent) relative to a specific acute angle – Sine, cosine and tangent ratios of acute angles in a right-angled triangle – Sines and cosines of complementary angles and their relationship – Application of trigonometric ratios to angles of elevation – Application of trigonometric ratios to angles of depression – Construction and use of a simple clinometer to measure angles of elevation and depression – Use of trigonometric tables and digital tools to find trigonometric ratios – Real-life applications of trigonometry in the immediate environment
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - identify the sides of a right-angled triangle in relation to a particular angle, - determine the sine, cosine and tangent of acute angles in a right-angled triangle, - relate sines and cosines of complementary angles, - apply trigonometric ratios to angles of elevation and depression, - recognise the use of trigonometry in real-life situations.
Core Competencies	– Self-efficacy: The learner makes observations and establishes the relationships between the sine and cosine ratios of complementary angles, building confidence in their own mathematical reasoning as they generate and verify the pattern independently. – Learning to learn: The learner learns collaboratively while constructing a simple clinometer using locally available materials (cardboard, straw, string, protractor) and using it as a hands-on learning resource for measuring angles of elevation and depression in the school environment.
Core Values	– Love: The learner demonstrates a caring attitude towards others when using a simple clinometer to measure the angle of elevation and depression of different objects in the immediate environment, ensuring every group member has a turn and that measurements are taken safely and considerately. – Responsibility: The learner takes care of instruments such as rulers, geometrical sets, graph paper and digital devices used during trigonometry activities, returning them in good condition and ensuring shared resources are available for peers.
Science & Engineering	– Asking Questions and Defining Problems: Learners articulate what information is needed (angle, known side) to find an

Practices	unknown height or distance, framing trigonometry as a tool for answering real measurement questions about their environment. – Planning and Carrying Out Investigations: Learners design and carry out a clinometer investigation — identifying objects to measure, deciding on measurement positions, recording angle and distance data, and checking results for consistency. – Using Mathematics and Computational Thinking: Learners select and apply the appropriate trigonometric ratio (sin, cos, tan) to compute unknown sides and angles, evaluate answers for reasonableness, and use digital tools (calculators, online resources) to verify computations. – Constructing Explanations: Learners use their generated table of complementary angles and their sine/cosine values to construct a reasoned explanation for why $\sin \theta = \cos(90^\circ - \theta)$, connecting numerical patterns to geometric reasoning. – Obtaining, Evaluating and Communicating Information: Learners present clinometer findings and trigonometric solutions to peers, critique each other's methods, and evaluate the accuracy of real-world measurements against calculated values.
Pertinent & Contemporary Issues (PCIs)	– Self-esteem: The learners acquire and strengthen self-esteem while designing and making a simple clinometer using locally available materials (cardboard, string, protractor, drinking straw) and successfully using it as a functional learning resource for measuring real angles in their school environment. – Life Skills (Critical Thinking and Problem Solving): Learners apply trigonometric ratios to authentic measurement problems — such as estimating the height of a flagpole, water tank, or school building in Nairobi — developing analytical and decision-making skills transferable to daily life. – Education for Sustainable Development: Learners use locally sourced, low-cost materials to construct the clinometer, reinforcing resourcefulness and environmental awareness while achieving a meaningful mathematical learning outcome.
Career Connections	– Surveying and Land Management: Trigonometric ratios are foundational tools for surveyors who map land parcels across Kenya's counties — from the Rift Valley to the Coast — measuring distances and elevations using clinometers and theodolites. – Architecture and Building Construction: Architects and construction engineers in Kenya use angles of elevation, roof pitch calculations and trigonometric ratios when designing and constructing buildings, ensuring structural stability. – Aviation and Navigation: Pilots and navigators at airports such as JKIA (Nairobi) and Moi International Airport (Mombasa) apply trigonometry to calculate flight approach angles, altitudes and distances. – Sports Coaching and Biomechanics: Sports scientists and coaches working with Kenyan athletes (e.g., track and field at Kipchoge Keino Stadium, Eldoret) use angle analysis to optimise throwing, jumping and running techniques. – Photography and Media Technology: Photographers and videographers computing camera angles, framing shots and calculating perspective use principles directly related to trigonometric ratios. – Forestry and Environmental Science: Forest officers in Kenya's Aberdare and Mau Forest complexes use clinometers to measure tree heights and terrain slopes — a direct application of angles of elevation.
Focus for Lessons	This unit introduces learners to trigonometry through a practical, investigation-led approach anchored in the Kenyan built and natural environment. Learners move from identifying the sides of a right-angled triangle, through deriving and applying the three trigonometric ratios, to discovering the complementary-angle relationship and finally deploying their knowledge to solve real problems involving angles of elevation and depression using a student-built clinometer. Throughout the 8 lessons, every concept is connected back to the anchoring phenomenon of estimating the height of an inaccessible structure using only a measured distance and an angle, reinforcing trigonometry as a practical, empowering tool rather than an abstract set of rules.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment — without ever measuring it directly? KICD KEY INQUIRY QUESTION:

	1. Why is trigonometry important?
Anchoring Phenomenon	ANCHORING PHENOMENON — "How tall is the water tank without climbing it?" At many Kenyan schools and homesteads, a large elevated water tank (sometimes mounted on a steel frame 4–8 metres high) is a familiar sight. Students can clearly see the tank from the ground but cannot climb it or use a tape measure along its height directly. The teacher presents this puzzling scenario: Two students stand at different positions on the ground, both a known horizontal distance from the base of the water-tank tower. Each student uses a simple angle-measuring device (a protractor and string) to sight the top of the tank and records a different angle. Without any additional climbing or measuring, can we use ONLY these two pieces of information — the horizontal distance and the measured angle — to calculate exactly how high the top of the tank is above the ground? This is puzzling because: (1) The students never measured the height directly, yet they can calculate it precisely. (2) Two students at different distances get different angles but should arrive at the SAME height — a powerful consistency check. (3) The same method, scaled up, is how surveyors map the Kenyan highlands, how engineers design the Thika Superhighway overpasses, and how foresters in the Mau Forest measure tree heights without felling them. The phenomenon is revisited and refined in every lesson: learners begin unable to explain it, gradually acquire the ratio tools (sin, cos, tan), discover the complementary-angle shortcut, build their own clinometer, and in the final lesson use the clinometer outdoors to measure a real structure and confirm their prediction — closing the explanatory loop completely.
Storyline Thread	Lesson 1 — PREDICT: "The Water Tank Problem" (Anchoring Phenomenon Launch) Learners observe a photograph/diagram of the school water tank or a similar elevated structure. In small groups they predict: "What information would you need, and how would you use it, to find the height of the tank without climbing?" Groups record predictions on sticky notes for the Driving Question Board (DQB). Teacher introduces vocabulary: right-angled triangle, hypotenuse, opposite side, adjacent side. Learners physically identify these sides in right-angled shapes around the classroom (door frames, window corners, book corners) with reference to a named acute angle. DQB is set up with the driving question posted prominently. Lesson 2 — OBSERVE (Part 1): Deriving the Three Trigonometric Ratios Learners use rulers and protractors to draw three different right-angled triangles each with the same acute angle (e.g., 35°) but different sizes. They measure the opposite, adjacent and hypotenuse for each triangle and compute the three ratios (opp/hyp, adj/hyp, opp/adj) in a table. Learners notice each ratio remains constant regardless of triangle size — a key discovery. Teacher names the ratios: sine, cosine, tangent. Learners use a scientific calculator to verify their measured ratios against calculator values. Connection to phenomenon: "These constant ratios are exactly what let us calculate the tank height from any distance!" Lesson 3 — OBSERVE (Part 2): Sine and Cosine of Complementary Angles Learners generate a table of angle pairs that sum to 90° (e.g., 20° & 70°, 35° & 55°, 45° & 45°) and record their sines and cosines. They discover that $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$. Learners construct an explanation for this pattern using a right-angled triangle diagram, noting that the "opposite" for one angle is the "adjacent" for its complement. DQB is updated: new understanding posted, new questions raised (e.g., "Does this work for ALL angles?").

	Lesson 4 — EXPLAIN (Part 1): Applying Trigonometric Ratios to Find Unknown Sides Guided practice: learners use $\sin$, $\cos$ and $\tan$ to calculate unknown sides in right-angled triangles drawn on graph paper. Problems are set in Kenyan contexts — length of a roof rafter in Kisumu, distance across a stream in Kericho, height of a mango tree in a Mombasa compound. Learners practise selecting the correct ratio, setting up the equation and solving. Peer checking of answers. Connection to phenomenon: learners set up and solve the water-tank calculation for the first time using a given angle and distance. Lesson 5 — OBSERVE (Part 3): Angle of Elevation — Clinometer Construction (Project Lesson) Learners construct a simple clinometer using a cardboard protractor, drinking straw, string and a small weight (stone or metal washer). Teacher models how to sight along the straw to an elevated object and read the angle of elevation. Learners practise sighting objects of known approximate height in the classroom. Discussion: what is the "angle of elevation"? How does it relate to the right-angled triangle? DQB updated with new vocabulary and connections. Lesson 6 — OBSERVE (Part 4): Angle of Depression and Outdoor Measurement Learners take their clinometers outdoors. Working in groups of 3–4, they measure the angle of elevation to the top of the school flagpole, water tank, or a tall tree, and record the horizontal distance from the base. They also identify and measure an angle of depression (e.g., sighting from a raised veranda down to a ground-level marker). Data is recorded on structured worksheets. Back in the classroom, each group calculates the height of the object using the appropriate trigonometric ratio. Groups compare results and discuss sources of error. Lesson 7 — EXPLAIN (Part 2): Problem Solving — Angles of Elevation and Depression Structured problem-solving lesson using multi-step real-life scenarios: a Rift Valley farmer estimating the height of a cliff face, a diver at Watamu estimating depth from a boat, a construction worker in Nairobi checking the angle of a ramp. Learners work individually then compare with a partner. Common errors (confusing opposite/adjacent, not identifying the correct reference angle) are addressed explicitly. DQB is updated: learners move sticky notes from "questions" to "answered" and add new refined understandings. Learners revisit their original predictions from Lesson 1 and evaluate them. Lesson 8 — MODEL BUILDING & SUMMATIVE CLOSURE: "Solving the Water Tank Problem" Learners use all accumulated knowledge to produce a final written and diagrammatic solution to the original anchoring phenomenon. Each learner draws a labelled diagram of the water-tank scenario, identifies the right-angled triangle, selects and applies the correct trigonometric ratio, calculates the height, and writes a clear explanation of their method. Groups present their clinometer project findings (angle, distance, calculated height, reflection on accuracy). Teacher facilitates a whole-class DQB review — all original questions are revisited and either answered or identified as leading to future learning (e.g., trigonometry of non-right-angled triangles). Formative assessment via class activity and portfolio entry.
Supporting Phenomena	– THE LEANING SIGN BOARD: A rectangular signboard on a tilted pole outside the school gate — learners notice the pole makes an acute angle with the ground. How do we find the actual height the sign reaches without measuring along the pole? (Supports: identifying opposite/adjacent/hypotenuse and setting up a $\tan$ ratio.) – THE SHADOW PUZZLE: At midday, a vertical maize stalk in the school garden casts a short shadow; in the late afternoon the same stalk casts a much longer shadow. Why does the shadow length change, and can we use the shadow to find the height of the stalk? (Supports: angle of elevation of the sun, $\tan$ ratio, and the idea that the same height gives different ratios at different angles.) – THE MIRROR ON THE FLOOR: A small mirror is placed flat on the ground between a learner and the base of a wall. The

- | | |
|--|---|
| | <p>learner steps back until they can just see the top of the wall in the mirror. The triangle formed is similar to the wall triangle. How is this related to the ratios we computed? (Supports: connecting similar triangles to constant trigonometric ratios.)</p> <ul style="list-style-type: none">– THE STAIRCASE RAMP: A newly built concrete ramp at a Nairobi hospital rises 1.2 m over a horizontal run of 4.8 m. Is this ramp accessible for wheelchair users (angle $< 14^\circ$)? (Supports: applying arctan/inverse trigonometry and angles of elevation in a real accessibility-design context.) |
|--|---|

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.

LESSON 1 (40 minutes (1 lesson); extend to a double lesson if the side-naming practice is not finished): The Water Tank Problem: Predict and Name the Sides

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon (how tall is the Kilimani Secondary School water tank without climbing it?), to record learners' own predictions about how it might be done, and to give learners the vocabulary they need to name the hypotenuse, opposite and adjacent sides of a right-angled triangle relative to a chosen acute angle θ .
Knowledge	By the end of the lesson the learner should be able to: (a) identify the right-angled triangle in relation to a particular angle, naming the hypotenuse, the opposite side and the adjacent side (KICD Specific Learning Outcome 2.3 a); (b) explain that the names opposite and adjacent change when the chosen angle changes, while the hypotenuse does not; (c) describe the tank problem as a right-angled triangle made by the tower, the ground and the line of sight.
Skills	NGSS-aligned practices: Asking Questions and Defining Problems (learners write questions for the Driving Question Board); Developing and Using Models (learners draw an initial model of the tank, ground and line of sight); Constructing Explanations (learners give a reasoned prediction). Mathematical processes: reasoning, communication, connection and collaboration. Core competencies: Self-efficacy (the learner makes observations and establishes relationships) and Learning to learn (the learner learns collaboratively).
Attitudes	Love: the learner demonstrates a caring attitude to others by listening and taking turns when working in pairs and sharing predictions. Self-esteem (PCI): the learner feels confident that every prediction is valued at the start of a new topic. Curiosity and appreciation of how mathematics is used in real life.
Key Inquiry Question	Why is trigonometry important? (KICD Key Inquiry Question). Driving question: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment, without ever measuring it directly?
Purpose in Storyline	This is the PREDICT lesson of the unit. Learners meet the water tank problem, make predictions without any new formula, and gain the side-naming language that Lesson 2 uses to build sine, cosine and tangent. The predictions and the Driving Question Board (DQB) created today are revisited in Lesson 7 (evaluating the predictions) and answered fully in Lesson 8. No numbers for the tank are given today and the tank's height is not revealed.
Safety Notes	This lesson is carried out entirely indoors or at the classroom door, and nobody is to climb the tank, its steel frame or any structure. If the class looks out at the tank tower, they stay on level ground, keep away from the tower base and do not look directly at the sun. Use blunt-ended pencils and keep rulers and protractors on the desks. Reinforce the message that the whole point of trigonometry is to measure things safely, without climbing.

B. LESSON OVERVIEW

Madam Achieng opens the lesson by describing the steel-framed water tank tower at Kilimani Secondary School in Nakuru. Everyone can see the tank but nobody may climb it, and a tape cannot be run up it from the ground. Two Grade 10 learners, Wanjiru and Otieno, stand at different distances from the base of the tower. Each sights the top of the tank with a simple protractor-and-string angle-measuring device, and they record different angles because they stand at different distances. Learners predict, individually and then in pairs, whether the height of the tank can be found from only a horizontal distance and an angle, and whether both learners should get the same answer. No numbers are given today. Learners sketch a first model of the situation and see that it forms a right-angled triangle, using what they already know from Grade 9 about right-angled triangles and Pythagoras. They then learn to name the hypotenuse, opposite and adjacent sides relative to a chosen angle θ , practising with the triangle of the tank, a door frame and a roof truss. Finally they open the Driving Question Board (DQB) with questions from their own wondering. The DQB has the

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Trigonometry Word Problems Practice Source: CK-12  Search ARES for similar readings	Learners look at a large sketch or a view from the window of the water tank tower at Kilimani Secondary School. They hear the story: Wanjiru stands near the tower and sights the top of the tank with a protractor and string, and Otieno stands further back and does the same. They are told that the two learners record different angles, and that no numbers are being given yet. On a sheet or in their exercise books, learners individually write: (1) How could we find the height of the top of the tank without climbing it or hanging a tape down from it? (2) Wanjiru is closer than Otieno. Who do you think records the larger angle? (3) Will Wanjiru and Otieno calculate the same height or different heights? Why? They then sketch a quick picture of the tank, the ground and the line of sight. Then they share their ideas with a partner and as a class. Learners may predict things such as 'use the shadow' or 'draw a scale diagram', which are welcome.	Madam Achieng says: "Look out of the window at our tank tower. Nobody is allowed to climb it. Today's problem is: how tall is the top of the tank, without climbing it?" She describes Wanjiru and Otieno and the two different angles, drawing a simple diagram on the board with two stick figures at different distances but no numbers. She says: "I will not tell you the answer or any formula today. I want your honest prediction." Then she gives 2 minutes of silent individual writing time. After the writing she gives WAIT TIME of 10-15 seconds before calling on anyone. She cold-calls 3 learners with: "Tell us your prediction and one reason." Then she takes a quick show of hands: "Who thinks Wanjiru gets the larger angle? Who thinks Otieno?" and records the counts on the board. She also asks, "Who thinks they will get the same height? Different?" She records the counts without judging them as right or wrong and keeps all the written predictions to revisit in Lesson 7.	Prediction with reasoning (Predict-Share-Revisit): writing a prediction and a reason commits learners to an idea, so they notice later when the evidence changes or confirms it. The class tally makes the disagreement about the larger angle and the same height visible, and creates a genuine need to know. It also surfaces preconceptions, such as thinking that the learner standing further away should see a larger angle.	Observe whether each learner writes a prediction with a reason and not just 'I don't know'. Indicators: at least 80 percent of learners have written an answer to all three questions; sketches show the tank, the ground and a sight line; the count of learners predicting the same height versus different heights is recorded. Note names of learners with misconceptions (for example, 'the farther learner gets a bigger angle') for follow-up in Lessons 2 and 5.
Observe Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)	In pairs, learners examine a labelled diagram of the tank problem prepared by the teacher on the board or on a printed sheet. It shows a vertical tower, level ground, a horizontal line from the learner to the base of the tower, and a line of sight from the	Madam Achieng says: "Look carefully. What shapes can you see in this picture? Talk to your partner for one minute." She gives WAIT TIME of 10-15 seconds after the pair talk, then cold-calls 2 pairs: "What shape did you find, and where is the right angle?" She	Noticing and Wondering with a model diagram: a labelled picture lets every learner look for structure before any new terms are given. Connecting to prior knowledge (Grade 9 right-angled triangles and Pythagoras) lets learners see what they already can and cannot do,	Listen for pairs who can identify the right angle at the base of the tower and describe the triangle as made by the tower, the ground and the line of sight. Indicators: learners correctly state that the longest side is opposite the right angle; at least 3 pairs name an

Search ARES for similar videos EXERCISE: Trigonometry Word Problems Practice Source: CK-12 Search ARES for similar readings	learner's eye to the top of the tank, with the angle of elevation marked at the learner's eye. They notice that the three lines form a right-angled triangle, with the right angle at the base of the tower. They recall Grade 9 knowledge: the triangle has a right angle, and Pythagoras' theorem links the three sides. They then look at the same triangle from two different learners' positions (Wanjiru's and Otieno's) and see two triangles with the same vertical side but different horizontal sides and different angles. Learners also look at other right-angled triangles in the classroom: a door frame, a window frame, a roof truss seen through the doorway, and a set square, to see right-angled triangles in their own surroundings.	then asks, "What do you remember from Grade 9 about right-angled triangles and Pythagoras?" and cold-calls 2 learners, recording on the board that $a^2 + b^2 = c^2$ connects the three sides, with the longest side opposite the right angle. She points out that Pythagoras needs two sides to find the third, but here the unknown is a height and the information is an angle and a distance. She says: "Pythagoras alone cannot help us yet. We have an angle. What else do we need?" She holds back from naming sine, cosine or tangent, because that is for Lesson 2. She then walks around and asks pairs, "Where else in this room can you find a right-angled triangle?"	which makes the gap, using an angle to find a side, clear and motivates the unit.	example of a right-angled triangle in the room; learners state that Pythagoras needs two sides but the situation gives a distance and an angle. Note pairs who do not recognise the right angle and give them a set square to check.
Explain Phase VIDEO: Trig challenge problem: trig values & side ratios <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Trigonometric ratios in right triangles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners now learn how to name the sides of a right-angled triangle relative to a chosen acute angle θ (theta). Madam Achieng marks the angle of elevation at Wanjiru's eye as θ. With the class, they name: the hypotenuse (the side opposite the right angle, always the longest), the opposite side (across from θ, which in the tank problem is the vertical side up the tower), and the adjacent side (next to θ but not the hypotenuse, which here is the horizontal distance along the ground). In pairs, learners label printed or board triangles, such as a door-frame triangle, a roof truss triangle and a ramp triangle. They then do the key step: the angle is moved to the other acute angle of the same triangle, and learners discover that opposite and adjacent swap, while the hypotenuse stays the same. They practise with at least four	Madam Achieng says: "We need a shared language. Let's call the acute angle we are interested in θ, pronounced theta." She draws the tank triangle and models: "First find the right angle. The side across from it is the hypotenuse. Next, stand at θ and ask: which side is across from me? That is the opposite. The side that is next to me, and is not the hypotenuse, is the adjacent." She uses the same colour for each side name on every diagram. She asks, "In our tank triangle, which side is the height of the tank above eye level? Which is the distance Wanjiru stands from the tower?" and gives WAIT TIME of 10-15 seconds, then cold-calls 3 learners. She then moves the angle to the top vertex and asks: "Now θ is here. Which side is opposite now? Which is adjacent?" She cold-calls 2 pairs. She gives each pair 4	Modelling then guided practice with a contrasting case: moving the angle in the same triangle shows that the names depend on the chosen angle, which prevents the common misconception that opposite and adjacent are fixed labels. Using varied orientations breaks the habit of reading 'opposite' as 'on the other side of the page'. Re-tying the labels to the phenomenon, so the terms have a purpose.	Check the labelled triangles. Success indicators: learners correctly label the hypotenuse in all 4 triangles (including the rotated one); at least 75 percent label opposite and adjacent correctly for the marked θ; learners can explain 'the hypotenuse stays the same but opposite and adjacent swap when we move θ'. Common errors to look out for: calling the vertical side 'opposite' regardless of θ; calling the longest-looking side the hypotenuse instead of the one across from the right angle. Ask learners making these errors, 'Point to the right angle first, then to θ.' Give a quick exit ticket with one triangle to label.

	triangles, labelling the three sides for the angle marked θ each time, and explain the reason for each label to a partner.	triangles in different orientations (including one turned upside down so learners do not rely on position) to label, and circulates asking, "How do you know that is the opposite side?" She does not introduce sin, cos or tan; if a learner asks, she says: "Good question. Keep it on the board for our next lesson." She then returns to the phenomenon: "For Wanjiru and Otieno, the height is the opposite side and the distance is the adjacent. Their angles are different. Why might that matter?"		
Driving Question Board (DQB) Creation  VIDEO: Trigonometry Word Problems Principles - Basic <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Right triangle trigonometry word problems <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners open the class Driving Question Board. Madam Achieng writes the Driving Question at the top: 'How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment, without ever measuring it directly?' and draws three columns: Questions, Answered and New understandings. Individually, learners write at least two questions on sticky notes or pieces of paper about the tank problem and the tools they might need (for example: Why do Wanjiru and Otieno get different angles? How can an angle help us find a height? Can we get the same answer from two places? How do surveyors in the Kenyan highlands do this? How do engineers on the Thika Superhighway use it? How do foresters in the Mau Forest measure trees?). Then in groups of four they sort their questions into themes (angles, the instrument, the method, the real world), choose their best one and stick it on the board under Questions. The class also adds today's learning, 'We can name	Madam Achieng says: "Scientists and surveyors begin with questions. What are you still wondering about the tank?" She gives 2 minutes of silent writing time, then 3 minutes of group sorting. She gives WAIT TIME of 10-15 seconds before asking for volunteers, then cold-calls 4 groups to read their chosen question aloud and say why it matters. She places each question on the DQB under a theme heading, grouping similar ones. She says: "Which of these can we start answering today? Which need more tools?" She moves 'What are opposite and adjacent sides?' into Answered, with a short note, and adds 'Side names depend on θ' under New understandings. She promises: "We will return to this board in every lesson." She points out the links between questions: for example, 'Why do different distances give different angles?' will be linked to Lessons 2 and 5. She leaves the board visible or photographs it for later lessons.	Driving Question Board: learners' own questions create ownership and a map of the unit. Sorting questions into themes helps learners see the structure of what they need to learn, and the Answered and New understandings columns give a visible record of growing understanding that the class returns to in every lesson.	Look at the quality of questions. Indicators: each learner has written at least 2 questions; at least half of the groups have a question that is linked to the mathematics (angle, distance, height, ratio) and not just a factual question; at least 1 question is about why the same height should come from different angles. Learners whose questions are only 'What is the answer?' should be asked, 'What would you need to know to work it out?' Keep a list of unanswered questions to guide the next lessons.

	hypotenuse, opposite and adjacent', to the New understandings column.			
Model Building Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Ordering triangle sides and angles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners draw an Initial Model in their exercise books, titled 'How can we find the height of the tank without climbing it?' The model includes: a labelled sketch of the tank tower, the ground, and Wanjiru's position; the line of sight to the top of the tank; a right-angled triangle drawn on the sketch with θ marked at the learner's eye and the three sides labelled hypotenuse, opposite and adjacent; a second position for Otieno further from the tower with its own angle marked; and a written sentence for their current idea: 'I think we can find the height by ...'. A short box beneath says 'What I am still not sure about'. They share the model with a partner and add one improvement based on the partner's comments. The model stays in their books and is revised in each lesson, with the final version completed in Lesson 8.	Madam Achieng says: "Draw your first model of the tank problem. Show the tower, the ground, the line of sight and the right-angled triangle. Label the sides for the angle θ at the learner's eye." She allows 6 minutes. She circulates and asks: "Where is the hypotenuse in your picture? What is the opposite side for the angle you marked?" She gives WAIT TIME of 10-15 seconds after each question. She then cold-calls 3 learners to explain their models and chooses one model to display. For the exit ticket she asks learners to write one sentence: 'The side that stands for the height of the tank above eye level is the _____ side.' She then closes: "Next lesson we will see whether the ratio of the sides changes when the triangle gets bigger or smaller while the angle stays the same. Keep your predictions safe. We will test them." She collects the exit tickets.	Initial model building: drawing a model before instruction records current thinking and shows learners that models are revised as evidence comes in. Including the labelled triangle and two positions turns the phenomenon into a mathematical structure, ready for the ratios in Lesson 2. The 'still not sure' box encourages honest uncertainty.	Review models and exit tickets. Indicators: the model shows a right-angled triangle with the right angle at the base of the tower; the vertical side to the top of the tank is labelled opposite and the ground distance is labelled adjacent relative to θ at the eye; two different positions are shown with different angles; the exit ticket answer is 'opposite'. Learners who label incorrectly are to be reminded to start with the right angle and then θ, and re-checked at the start of Lesson 2.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners make real predictions about the tank, and what proportion predicted that Wanjiru and Otieno would get the same height? Keep the tally for Lesson 7. (2) Which side-naming errors were most common (for example, calling the vertical side always opposite, or mislabelling the hypotenuse when the triangle was rotated)? Plan quick re-practice at the start of Lesson 2. (3) Were the DQB questions about the mathematics, and which questions can be connected directly to Lessons 2 to 8? (4) Did every learner, including quieter ones, take part in sharing predictions? (5) Did you resist naming sine, cosine and tangent today? Note any learners who already know the terms and could help others in Lesson 2. (6) Is the DQB placed where learners can see it, and are the sticky notes secure? Note changes for the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the water tank tower at Kilimani Secondary School cannot be climbed, yet Wanjiru and Otieno stand at different
---------------------	---

	distances from its base and sight the top of the tank to record different angles. The tower, the ground and the line of sight form a right-angled triangle.
What did I learn?	We learned to name the sides of a right-angled triangle relative to a chosen acute angle θ : the hypotenuse is opposite the right angle and is the longest side; the opposite side is across from θ ; the adjacent side is next to θ but is not the hypotenuse. When θ moves to the other acute angle, opposite and adjacent swap but the hypotenuse stays the same.
How does this explain the phenomenon?	We can say that the height of the tank above eye level is the opposite side and the distance from the learner to the tower is the adjacent side for the angle at the learner's eye, but we cannot yet explain how the angle and distance give the height, or whether Wanjiru and Otieno should get the same answer. These questions are on our Driving Question Board and we will answer them in the next lessons.

LESSON 2 (80 minutes (2 x 40-minute lessons)): Deriving sine, cosine and tangent: Why does the tank method work at any distance?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners discover that for a fixed angle the ratios of the sides of a right-angled triangle stay the same however big the triangle is, and name these ratios sine, cosine and tangent. This explains why Wanjiru and Otieno can stand at different distances and still find the tank height.
Knowledge	By the end of the lesson the learner should be able to: a) identify the right-angled triangle in relation to a particular angle; b) determine the sine, cosine and tangent of acute angles in a right-angled triangle. Definitions: $\sin \theta = \text{opposite} \div \text{hypotenuse}$, $\cos \theta = \text{adjacent} \div \text{hypotenuse}$, $\tan \theta = \text{opposite} \div \text{adjacent}$.
Skills	Measuring lengths to the nearest 0.1 cm; calculating ratios to 3 decimal places; analysing and interpreting data (NGSS SEP: Analyzing and Interpreting Data); looking for patterns and constructing explanations from evidence; using a calculator in DEG mode.
Attitudes	Core competency, Self-efficacy: the learner makes observations and establishes relationships between the ratios. Value, Love: the learner shows care for peers by sharing measuring tools and checking each other's work. Appreciate the use of trigonometry in real-life situations. PCI, Self-esteem: learners gain confidence by discovering a rule themselves from their own data.
Key Inquiry Question	Why is trigonometry important? (KICD) Lesson question: Why do Wanjiru, at 8 m, and Otieno, at 10 m, get different angles but the same tank height, and what is it about a triangle that stays the same?
Purpose in Storyline	Lesson 1 launched the tank problem and named the sides opposite, adjacent and hypotenuse. Lesson 2 supplies the key idea behind the method: for a fixed angle the side ratios do not depend on triangle size, so they can be named (sin, cos, tan) and read from a calculator. Later lessons use this to find unknown sides, discover the complementary angle link and build the clinometer.
Safety Notes	Sharp pencils and compass points: carry them point-down and do not point them at others. Use rulers sensibly, not as swords. Keep the walkway clear when groups move to the DQB. Learners share one calculator per pair and handle it carefully.

B. LESSON OVERVIEW

Madam Achieng opens by recalling the Kilimani Secondary School tank tower from Lesson 1: Wanjiru stands 8 m away and reads a bigger angle than Otieno, who stands 10 m away. Both somehow get the same height. Learners predict why the triangles formed can give the same answer. Groups then measure three printed right-angled triangles (A, B and C) that all have a 35° angle but different sizes, and record adjacent, opposite and hypotenuse in cm. Triangle A has sides 4.0, 2.8 and 4.9 cm; B has 8.0, 5.6 and 9.8 cm; C has 12.0, 8.4 and 14.7 cm. Learners calculate opposite/hypotenuse, adjacent/hypotenuse and opposite/adjacent for each triangle and find that every triangle gives about 0.571, 0.816 and 0.700 (real measurements vary by about ± 0.02). In the Explain phase the teacher names the ratios sine, cosine and tangent (sin, cos, tan afterwards), and learners check on a calculator in DEG mode that $\sin 35^\circ = 0.5736$, $\cos 35^\circ = 0.8192$ and $\tan 35^\circ = 0.7002$. They connect the constant ratio to the tank: the ratio opposite $\div$ adjacent depends only on the angle, not on the triangle size, which is why 8 m with 37° and 10 m with 31° can lead to one height. The DQB is updated and the class tank model is revised. Learners do not yet calculate the tank height; that is Lesson 4.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Intro to the trigonometric ratios Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look at the Lesson 1 diagram of the tank tower with Wanjiru (8 m) and Otieno (10 m) at different distances. Madam Achieng draws two right-angled triangles on the board, one small and one large, with the same angle at the bottom. In pairs learners individually write a prediction: 'If two right-angled triangles have the same angle but different sizes, what do you think stays the same and what changes?' They sketch the two triangles and mark opposite, adjacent and hypotenuse relative to the angle (building on Lesson 1). They share ideas with a partner and write their best prediction on a sticky note: 'I think ____ stays the same because ____.'	Say: 'Remember Wanjiru and Otieno at Kilimani Secondary School in Nakuru. Wanjiru is closer and gets a bigger angle. Yet both find the same tank height. Today we find out what stays the same.' Draw the two triangles with a 35° angle. Ask: 'Which sides get bigger when the triangle gets bigger? Which angle does not change?' Give 60 seconds of silent individual thinking, then pair talk for 2 minutes. WAIT TIME: after asking 'What do you think stays the same?', wait 10-15 seconds before taking answers. Cold-call 4 learners to share predictions, including at least 2 who have not spoken yet, and say: 'Thank you, I am not saying right or wrong yet. We will test it with evidence.' Record the predictions on the board under 'Our predictions'. Circulate and note any learner who says 'the ratio' unprompted.	Prediction with a sketch (Developing and Using Models): learners commit to an idea before seeing data so they can later compare their reasoning with evidence. Writing the 'because' part makes their prior thinking visible and creates a reason to measure.	Observable indicators: (1) Learner sketches show correctly labelled opposite, adjacent and hypotenuse relative to the marked angle (Lesson 1 skill). (2) Learner states that the angle stays the same. (3) Tally the predictions: sides stay the same, a relationship between sides stays the same, or nothing stays the same. If more than a third mislabel the sides, spend 3 minutes re-labelling a triangle with the door frame before the Observe phase.
Observe Phase  VIDEO: Intro to the trigonometric ratios Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Trigonometric ratios in right triangles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners receive printed right-angled triangles A, B and C, all with a 35° angle and with the right angle at the bottom right. Each learner has a ruler and a pencil. They measure adjacent, opposite and hypotenuse for each triangle to the nearest 0.1 cm and record the lengths in a table. The expected ideal values are: A 4.0, 2.8, 4.9 cm; B 8.0, 5.6, 9.8 cm; C 12.0, 8.4, 14.7 cm. Next they use a calculator to work out $\text{opposite} \div \text{hypotenuse}$, $\text{adjacent} \div \text{hypotenuse}$ and $\text{opposite} \div \text{adjacent}$ for each triangle to 3 decimal places. Expected ratios: 0.571, 0.816 and 0.700 for every triangle, with real learner measurements varying by about	Distribute materials, with 2 learners sharing a ruler. Say: 'Label the 35° angle first. Then name each side from that angle: opposite, adjacent, hypotenuse.' Model the first measurement for Triangle A on the board with 'The adjacent side is next to the 35° angle, and it is not the hypotenuse.' Assign roles: measurer, recorder, calculator operator, checker, rotating every triangle. Circulate with these questions: 'Which side is the longest? Does your hypotenuse look longest?' and 'Measure from the corner, not from the ruler's edge.' Check pairs: if a group gets ratios differing by more than 0.03, ask them to re-measure the side	Data table pattern-finding (Analyzing and Interpreting Data): learners generate their own data and compare columns to find an invariant. The class pooled table shows that small measurement errors do not hide the pattern, which builds trust in the evidence.	Observable indicators: (1) Lengths are measured to 0.1 cm and sides are labelled correctly from the 35° angle. (2) Ratios are computed correctly and in the expected range (about 0.57, 0.82, 0.70 within ± 0.02). (3) Learners state that the ratios are about the same even though the triangles differ in size. Check each group's table with a quick tick and watch for learners dividing in the wrong order. Note any group still believing the ratio grows with the triangle, and revisit its table with it.

	±0.02. Groups then compare the ratio columns across the three triangles and write what they notice in one sentence.	that seems off. After the ratio calculations say: 'Put your pencils down. Look at the three columns. What do you notice?' WAIT TIME: wait 10-15 seconds in silence. Then cold-call 3 groups to read out their Triangle A, B and C ratios for one column and record them on a class table on the board. Ask: 'Triangle C is three times as big as Triangle A. Did the ratio become three times as big?'		
Explain Phase  VIDEO: Finding reciprocal trig ratios <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Evaluate inverse trig functions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners explain in their groups why the ratios stayed the same: the three triangles are enlargements of one another with the same angles, so the sides are in proportion (linking to similarity from Strand 2.1). Madam Achieng then gives the names: the ratio opposite ÷ hypotenuse is called the sine of the angle (sin), adjacent ÷ hypotenuse is the cosine (cos), and opposite ÷ adjacent is the tangent (tan). Learners write the definitions in their notebooks: $\sin \theta = \text{opposite} \div \text{hypotenuse}$, $\cos \theta = \text{adjacent} \div \text{hypotenuse}$, $\tan \theta = \text{opposite} \div \text{adjacent}$. They set their calculators to DEG mode and check $\sin 35^\circ = 0.5736$, $\cos 35^\circ = 0.8192$ and $\tan 35^\circ = 0.7002$ against their class table, and discuss why the calculator values are slightly different from the hand-measured 0.571, 0.816 and 0.700 (measuring with a ruler to 0.1 cm has small errors). In pairs they then answer: 'How does this explain Wanjiru and Otieno getting the same tank height?' The expected idea is that tan of the angle gives the ratio of the height above eye level to the distance, and the angle tells us that ratio regardless of how far away the	Say: 'Why might the ratios stay the same? Think about similar shapes: if a triangle is enlarged, what happens to all its sides?' WAIT TIME: wait 10-15 seconds, then cold-call 3 learners. Introduce the names: 'Mathematicians gave these three ratios names. Opposite over hypotenuse is the sine. Adjacent over hypotenuse is the cosine. Opposite over adjacent is the tangent. We write sin, cos and tan.' Write the three definitions on the board. Say: 'Switch your calculator to DEG mode. Type sin 35 and read me the answer.' Check every pair's screen, because RAD mode is a common error. Ask: 'Our hand answer was 0.571 and the calculator says 0.5736. Why are they close but not identical?' Cold-call 2 learners. Then link to the phenomenon: 'Wanjiru reads 37° and Otieno reads 31°. The tangent changes with the angle, but for any one angle it does not change with the size of the triangle. How could that help them both find the same height?' Give 2 minutes of pair talk and cold-call 3 pairs. Confirm: 'Next lesson we use these to find sides, and Lesson 4 gives the full tank calculation.' Do not use SOH-CAH-TOA yet beyond a brief	Claim-Evidence-Reasoning (Constructing Explanations): learners state the claim (the ratios depend only on the angle), cite their table as evidence, and reason with similarity (enlargement keeps the sides in proportion). Naming the ratios after the discovery gives the vocabulary meaning.	Observable indicators: (1) Learners write sin, cos and tan definitions correctly in terms of opposite, adjacent and hypotenuse. (2) Calculator in DEG mode gives 0.5736, 0.8192 and 0.7002 for 35°. (3) Learners explain that the ratio stays the same because the triangles are similar or in proportion. (4) Learners link the idea to the tank in a sentence such as 'The angle fixes the ratio, so the distance does not change the answer.' Use a 3-question exit check: write the formula for $\tan \theta$; state $\sin 35^\circ$ to 4 decimal places; state what stays the same when a triangle is enlarged. Common errors to look for: inverted ratios and RAD mode.

	person stands. They do not yet calculate the height.	mention if learners ask.		
Driving Question Board (DQB) Creation  VIDEO: Solving for a side in right triangles with trigonometry <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Right triangle trigonometry word problems <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the Driving Question Board with the driving question: 'How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment, without ever measuring it directly?' The board has three columns: 'Questions', 'Answered' and 'New understandings'. Groups review the Lesson 1 sticky notes and move any question now answered, such as 'Why do different angles give the same height?' (partly answered), to 'Answered' or add a note. Each group writes at least one 'New understanding' on a sticky note, e.g. 'For 35°, opposite ÷ adjacent is always about 0.70' or 'Ratios stay the same for any size of triangle at the same angle.' They also add one new question, e.g. 'How do we use tan to find the height of the tank?' or 'Why do sin and cos use the hypotenuse?'	Say: 'Walk to the board with your group. Read the questions from Lesson 1. Which can we now answer, even partly?' Allow 3 minutes. Ask a group to read a note aloud and ask: 'What evidence from today's table supports moving it?' Press for evidence rather than opinions. WAIT TIME: wait 10-15 seconds after asking 'What new question do we have?' and cold-call 3 groups to read their new question. Sort the new questions by theme: using the ratios to find a side, the angle of elevation, building a clinometer. Say: 'Keep your question; we will answer it in the coming lessons.' Make sure that quieter learners also post notes by asking each group to name a different writer than last time.	Public question tracking (Asking Questions and Defining Problems): the DQB makes the growth in understanding visible. Moving notes between columns requires learners to cite evidence, and new questions point to the next lessons.	Observable indicators: (1) Each group moves or annotates at least one note with a piece of evidence from the triangle table. (2) New understanding notes state the constant-ratio idea in the learner's own words. (3) New questions are specific and testable (e.g. about using tan to find height). Look for groups still posting only vague questions and prompt them with 'Which triangle side would you know and which would you want?'
Model Building Phase  VIDEO: Sine & cosine of complementary angles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Trig values of	Learners revise the class tank model begun in Lesson 1. On a large chart, the class redraws the tank tower diagram with Wanjiru at 8 m and Otieno at 10 m, showing two right-angled triangles with different angles but both reaching the same height. Beside the diagram, learners add a 'Ratio box' with the definitions of sin, cos and tan and the calculator values for 35°. They add a sentence in their own notebooks: 'The ratio opposite ÷ adjacent depends only on the angle, not on the size of the triangle, so it can be used to find a height from a distance.' Each	Display the Lesson 1 chart and say: 'Our model last time had the tank, the two learners and the side names. What do we add today?' Draw the diagram with student input, and ask: 'Where is the angle? Which side is opposite? Which is adjacent?' Add the ratio box as learners dictate it. Ask for an updated explanation: 'Why do both learners get the same height?' WAIT TIME: wait 10-15 seconds, then cold-call 3 learners to give a part of the explanation, each adding to the previous. Be careful not to compute the height; say: 'We do not yet calculate. That	Cumulative model revision (Developing and Using Models): learners add the new idea of a constant ratio to the shared tank model, linking the explanation to the phenomenon and showing what remains unexplained. The 'What I still need to know' step keeps the driving question open.	Observable indicators: (1) The revised diagram shows both triangles with opposite and adjacent sides labelled and the angle marked. (2) Learners' sentences state that the ratio depends on the angle and not on triangle size. (3) Learners avoid claiming that the angle itself changes with triangle size. (4) The 'still need to know' note identifies using the ratio to find an unknown side or measuring the angle. Check the Observed/Learned/Explained summary tables to decide who needs a short reteach of the three

special angles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	learner then updates their personal model by adding one thing that changed since the Predict phase, and writes a one-line answer to: 'What do I still need to know to find the tank height?' Expected answers: how to use the ratio with a known side, and how to measure the angle.	is Lesson 4, when we choose the right ratio and solve.' Hand out the summary table (Observed, Learned, Explained) and have learners complete it individually in 4 minutes. Collect it or check it at the door. Close with: 'Next time we explore a pattern between sine and cosine for different angles.'	ratios in the next lesson.
---	--	---	----------------------------

D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners discover the constant ratio from their own measurements, or did they rely on my statement? Which groups needed prompting? (2) Which sides did learners mislabel, and did this come from confusion between adjacent and hypotenuse? (3) How many learners had their calculator in RAD mode, and what routine will I use to check DEG mode in future lessons? (4) Did the small differences between hand-measured ratios (about 0.571, 0.816, 0.700) and calculator values (0.5736, 0.8192, 0.7002) support or confuse the idea of a constant ratio? (5) Could learners connect the constant ratio to Wanjiru and Otieno getting the same height? (6) Which learners have not yet spoken in a whole-class discussion, and how will I include them next time? (7) What should be adjusted for Lesson 3 on complementary angles?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did you measure and calculate for the three 35° triangles A, B and C? Record the adjacent, opposite and hypotenuse lengths, and the three ratios (opposite ÷ hypotenuse, adjacent ÷ hypotenuse, opposite ÷ adjacent) for each triangle. What did you notice about the ratios as the triangles got bigger?
What did I learn?	What are $\sin \theta$, $\cos \theta$ and $\tan \theta$ in terms of opposite, adjacent and hypotenuse? What are the calculator values of $\sin 35^\circ$, $\cos 35^\circ$ and $\tan 35^\circ$ in DEG mode, and how close are they to your measured ratios?
How does this explain the phenomenon?	Why do the ratios stay the same for triangles of different sizes with the same angle? How does this idea help explain how Wanjiru (8 m, 37°) and Otieno (10 m, 31°) at Kilimani Secondary School can both find the same tank height without climbing the tower? What do you still need to learn to calculate that height?

LESSON 3 (80 minutes (2 x 40-minute lessons)): Complementary Angles: Why Does $\sin 20^\circ$ Equal $\cos 70^\circ$?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To discover and explain why the sine of an acute angle equals the cosine of its complement, so that learners can choose between sin and cos with confidence when they use the ratios to find the height of the Kilimani Secondary School water tank.
Knowledge	By the end of the lesson the learner should be able to: (c) relate sines and cosines of complementary angles. Learners will state that $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$, and explain that the side opposite one acute angle is the side adjacent to the other acute angle.
Skills	Using the Lesson 2 definitions of sin and cos to calculate ratios in a right-angled triangle; using a calculator in DEG mode to generate a table of complementary angles; identifying patterns in data (SEP: Analyzing and Interpreting Data); constructing an explanation from a labelled diagram (SEP: Constructing Explanations); making generalisations and conclusions from a table.
Attitudes	Self-efficacy: the learner makes observations and establishes the relationships between the sine and cosine ratios. Learning to learn: the learner learns collaboratively with peers. Love: the learner shows care for others by listening and sharing the calculator in turn.
Key Inquiry Question	Why is trigonometry important?
Purpose in Storyline	Lesson 2 gave learners the definitions of sin, cos and tan and showed that the ratios depend only on the angle. Lesson 3 asks: when the angle is the other acute angle of the triangle, how do the ratios change? The answer, that sin and cos swap, means learners need fewer tools, and it checks the later use of the tank triangle from either acute angle. Lesson 4 will use the ratios to solve for unknown sides, including the tank-height calculation for Wanjiru (8 m, 37°) and Otieno (10 m, 31°).
Safety Notes	Keep the calculator on the desk and share it in turn. Use rulers and set squares carefully and do not point them at others. Keep bags and walkways clear while learners move to the board to add entries.

B. LESSON OVERVIEW

Madam Achieng reminds the class of the puzzle at Kilimani Secondary School, Nakuru: Wanjiru (8 m away) and Otieno (10 m away) sight the top of the tank tower and read different angles, yet the tank height must come out the same. In Lesson 2 learners found that sin, cos and tan depend on the angle, not on the size of the triangle. Today they notice that every right-angled triangle has two acute angles, and ask what happens to the ratios if we look from the OTHER acute angle. Learners first predict, then fill in a table of complementary pairs (20° and 70° , 30° and 60° , 35° and 55° , 45° and 45°) using a calculator in DEG mode, and then draw and label one triangle twice to explain why $\sin \theta = \cos(90^\circ - \theta)$. They add questions and answers to the Driving Question Board and revise the class model of how the tank height can be found from an angle and a distance.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Sine & cosine of complementary	(Approx. 12 minutes) Learners look at the Lesson 1 sketch of the tank tower on the wall. In pairs they draw a right-angled triangle	Say: 'Remember Wanjiru at Kilimani Secondary School. Every right-angled triangle has TWO acute angles. We have only looked	Predict-Share-Revise (Think-Pair-Share): writing a prediction with a reason before any data are seen exposes learners' current ideas	Look for: (1) learners who correctly state the other acute angle as 90° minus the first (53° for 37°); (2) learners who write a reason rather

angles Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Sine and Cosine of Complementary Angles (Flexbook) Source: CK-12 Search ARES for similar readings	for Wanjiru (the tower is the vertical side, the ground is the horizontal side, her line of sight is the hypotenuse). They mark her angle of elevation at her eye and then mark the OTHER acute angle at the top. They write: (1) How big is the other acute angle if one is 37°? (2) If Wanjiru's angle is 37°, what do you predict sin and cos of the OTHER angle will be compared with sin 37° and cos 37°? They write the prediction in their books with a reason, using a sentence starter: 'I think the sin of the other angle will be ____ because ____.'	at one. Today we look at the other.' Draw the triangle with the 90° corner and ask: 'If the three angles add to 180° and one is 90°, what do the two acute angles add up to?' WAIT TIME 10 seconds, then cold-call 2 learners. Introduce the word complementary: two angles that add to 90°. Give 3 minutes of silent individual thinking, then 3 minutes of pair talk. Say: 'I am not looking for the right answer yet. I want your best guess and your reason.' Cold-call 4 pairs with the sentence starter and record 4 predictions on the board without judging them. Do not confirm or correct.	(for example 'the ratios will stay the same' or 'they will be opposite') so that the later table can confirm or challenge each idea. This keeps the phenomenon of the tank triangle in view.	than just a guess; (3) learners who mark the top angle at the correct vertex. Walk round and tick on a class list who gave a reason. If more than a third give 53° wrongly, redraw the angle sum on the board before moving on.
Observe Phase  VIDEO: Trig word problem: complementary angles Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Sine and Cosine of Complementary Angles Practice Source: CK-12 Search ARES for similar readings	(Approx. 25 minutes) In groups of 4 with one calculator set to DEG mode per pair, learners complete the Lesson 3 table. For each pair of complementary angles they find the sin and cos to 4 decimal places: 20° and 70° (sin $20^\circ = 0.3420$, cos $20^\circ = 0.9397$, sin $70^\circ = 0.9397$, cos $70^\circ = 0.3420$); 30° and 60° (sin $30^\circ = 0.5000$, cos $30^\circ = 0.8660$, sin $60^\circ = 0.8660$, cos $60^\circ = 0.5000$); 35° and 55° (sin $35^\circ = 0.5736$, cos $35^\circ = 0.8192$, sin $55^\circ = 0.8192$, cos $55^\circ = 0.5736$); 45° and 45° (sin $45^\circ = \cos 45^\circ = 0.7071$). Rows are shared among group members and the results are pooled. Learners then circle values that match and write what they notice. They check that the 35° row agrees with the calculator value they used in Lesson 2 (sin $35^\circ = 0.5736$, cos $35^\circ = 0.8192$). Extension: they predict sin 80° from the table of cos 10° and test it.	Before starting, say: 'Check the top of your calculator screen for D or DEG. If it shows R or G, change it now. That is the commonest mistake.' Walk round and check 3 calculators in each row of desks. Say: 'Fill the table, then circle numbers that are equal. Do not explain yet.' Ask: 'What do you notice when you look across the row for 20° and 70°?' WAIT TIME 10 seconds, then cold-call 3 learners to read out what they circled. Ask: 'Is it a coincidence for 20° and 70°? What about 30° and 60°?' Cold-call 2 more groups to report their other rows. For groups finishing early, ask: 'Why do you think 45° gives the same value for sin and cos?' Do not give the reason yet. Collect findings on the class chart: 'sin of one angle = cos of the other'.	Pattern Finding in Data (Analyzing and Interpreting Data): learners generate a table from the calculator, look for repeated values across rows and compare predictions with data. Using four different angle pairs shows that the pattern is not a coincidence, and the 45° case acts as a special check.	Look for: (1) calculators in DEG mode with correct values to 4 d.p.; (2) learners circling sin 20° with cos 70° (and not sin 20° with sin 70°); (3) learners checking that the two angles in each row add to 90°; (4) a statement such as 'the sine of one angle equals the cosine of the other'. Record on a tally sheet which groups noticed the pattern unprompted. Listen for groups who say '45° is the same both ways' and invite them to share later.
Explain Phase  VIDEO:	(Approx. 25 minutes) Learners explain WHY the pattern occurs.	Say: 'Your table shows WHAT happens. Now we explain WHY.'	Claim-Evidence-Reasoning (Constructing Explanations): the	Look for: (1) a triangle with the opposite and adjacent sides

Sine & cosine of complementary angles Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Sine and Cosine of Complementary Angles (Flexbook) Source: CK-12 Search ARES for similar readings	Each learner draws a right-angled triangle with a 35° angle at A and 55° at B, with sides labelled a (opposite A), b (opposite B) and c (hypotenuse). They complete two tables: relative to angle A (35°) the opposite side is a and the adjacent side is b; relative to angle B (55°) the opposite side is b and the adjacent side is a. They write $\sin 35^\circ = a/c$ and $\cos 55^\circ = a/c$, then $\cos 35^\circ = b/c$ and $\sin 55^\circ = b/c$. They state the rule in their own words, then write it with symbols: $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$. They test it with Wanjiru's tank triangle: her angle is 37°, so the angle at the top of the sight line is 53°, which gives $\sin 37^\circ = \cos 53^\circ$ and $\cos 37^\circ = \sin 53^\circ$. They use the rule to answer: if $\sin 25^\circ$ is known, what is $\cos 65^\circ$? Why is $\sin 45^\circ = \cos 45^\circ$? Finally they revisit their predictions and mark them 'supported' or 'revise'.	Use your triangle labels.' Model the first label on the board, then let learners do the second. Ask: 'The side opposite 35° is the side a. Which side is next to 55° but not the hypotenuse?' WAIT TIME 10 seconds, cold-call 3 learners. Ask: 'So what happens to the opposite and adjacent sides when we change from the 35° angle to the 55° angle?' WAIT TIME 15 seconds, cold-call 2 learners to say it in a full sentence: 'The opposite side becomes the adjacent side.' Then ask: 'Why does 45° give equal sin and cos?' Guide: the two acute angles are both 45°, so the opposite and adjacent sides are the same length. Press for reasoning with 'How do you know?' and 'Show me on the diagram.' Address a common error: some learners write $\sin 20^\circ = \cos 20^\circ$. Ask them to check on the calculator. Invite learners to compare their prediction from the start and say whether it was supported. Remind them not to use the ratios to find sides yet; that is Lesson 4.	claim is '$\sin \theta = \cos(90^\circ - \theta)$', the evidence is the table from the calculator, and the reasoning is the swap of opposite and adjacent sides in the labelled triangle. Linking the number pattern to the diagram builds a causal explanation, not just a rule to memorise.	correctly relabelled for each angle; (2) a full-sentence reason that uses the words opposite, adjacent and hypotenuse; (3) the rule written correctly with $90^\circ - \theta$; (4) a correct answer for $\cos 65^\circ$ (equals $\sin 25^\circ$) and for the 45° case. Use a quick exit check: 'Write $\cos 40^\circ$ as a sine.' Expected answer: $\sin 50^\circ$. Note learners who answer $\sin 40^\circ$ for follow-up in the next lesson.
Driving Question Board (DQB) Creation  VIDEO: Trig word problem: complementary angles Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Sine and Cosine of	(Approx. 8 minutes) Learners go to the Driving Question Board, which has three columns: Questions, Answered and New understandings. They write on sticky notes any new questions the lesson raised (for example 'Can we use the 53° angle at the top of the tank instead of 37°?' or 'Does the same swap work for tan?' or 'How do we find the actual height using these ratios?'). They move any questions that are now answered from Questions to Answered (for example 'Do sin and cos always depend on the angle only?' from Lesson 2) and add a	Say: 'Come to the DQB in your groups. Each group adds at least one question and one new understanding.' Ask: 'Which question is Lesson 3 helping us to answer? Which is still waiting?' WAIT TIME 10 seconds. Cold-call 3 groups to read a sticky note aloud. Help the class to see that 'How tall is the tank?' is still in the Questions column and that the next lesson will use the ratios to find unknown sides. Praise questions that use the vocabulary opposite, adjacent and complementary. If no one asks about tan, prompt: 'We have only	Question Tracking on the DQB: moving questions to Answered and writing New understandings makes the growth of knowledge visible and keeps the class focused on the tank phenomenon. It also uses learners' own questions to set the direction for the next lesson.	Look for: (1) questions that connect to the tank (heights, other angle, using sin or cos to find lengths); (2) correct wording in New understandings, using complementary and opposite/adjacent; (3) at least one question from each group. Note any recurring misconception, such as 'sin and cos are the same', to address at the start of Lesson 4.

Complementary Angles Practice Source: CK-12 Search ARES for similar readings	note in New understandings: 'sin of an angle = cos of its complement because opposite and adjacent swap.'	looked at sin and cos. What about tan?' and add it as a class question without answering it.		
Model Building Phase  VIDEO: Finding reciprocal trig ratios <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Trigonometric ratios in right triangles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	(Approx. 10 minutes) Learners open the class model of the tank problem started in Lesson 1 and updated in Lesson 2. In pairs they add a new part: on the tank diagram (a vertical tower of unknown height, Wanjiru standing 8 m away with a 37° angle of elevation) they mark the second acute angle at the top of the sight line as 53° and write beside it 'sin $37^\circ = \cos 53^\circ$, $\cos 37^\circ = \sin 53^\circ$ '. Under the diagram they write a new rule for the model: 'Either acute angle can be used. Sin and cos swap when we move to the complementary angle.' They do the same quickly for Otieno (10 m, 31°) and mark his complement as 59° . They write a new sentence about what they still need: 'We know the ratios depend only on the angle, and that sin and cos are linked. We still need a way to use a ratio to find the tank height.' Then they complete the exit ticket: 'Write $\cos 70^\circ$ as a sine and explain in one sentence why it works.'	Say: 'Open the model. What has changed since yesterday? Mark in a different colour pen what we added today.' Ask: 'Look at Otieno's triangle. What is his other acute angle? How do you know?' WAIT TIME 10 seconds, cold-call 3 learners. Prompt: 'Does this solve the height? What is still missing?' Cold-call 2 learners to say what they still need. Make the link: 'Next lesson you will choose the correct ratio from the side you know and the side you want, and solve for the unknown side. That is how Wanjiru and Otieno will find the same tank height.' Collect the exit tickets at the door. Do not reveal the height of the tank.	Cumulative Model Revision (Developing and Using Models): learners add the complementary-angle relationship to a diagram they have already built, so that the model of the tank grows lesson by lesson. Identifying what is still missing prepares them for the use of ratios to find unknown sides in Lesson 4.	Look for: (1) both acute angles marked correctly on each learner's triangle (53° and 59°); (2) the correct rule written; (3) an explicit statement of what is still needed (a way to find the unknown side). On the exit ticket, check that $\cos 70^\circ$ is written as $\sin 20^\circ$ with a reason that mentions opposite and adjacent swapping. Sort the tickets into correct, partly correct (right answer, no reason) and incorrect to plan the first five minutes of Lesson 4.

D. TEACHER REFLECTION

After the lesson ask yourself: (1) Did learners explain the pattern using the swap of opposite and adjacent sides, or did they only copy the rule $\sin \theta = \cos(90^\circ - \theta)$? (2) Which learners confused $\sin 20^\circ = \cos 70^\circ$ with $\sin 20^\circ = \sin 70^\circ$, and what diagram will help them? (3) Were all calculators in DEG mode, and did any learners need support with the buttons? (4) Did the wait time give quieter learners, including girls and learners who speak mostly Kiswahili or mother tongue, space to share? (5) How well did learners connect the lesson to Wanjiru and Otieno at Kilimani Secondary School? Note which exit tickets need follow-up before Lesson 4, where learners choose a ratio and solve for an unknown side. Do not reveal the tank height, since the class will work towards it in Lessons 4 to 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Complete the Lesson 3 table. For the pairs $20^\circ/70^\circ$, $30^\circ/60^\circ$, $35^\circ/55^\circ$ and $45^\circ/45^\circ$, we found that sin of one angle equals cos of the other (for example $\sin 20^\circ = \cos 70^\circ = 0.3420$ and $\sin 35^\circ = \cos 55^\circ = 0.5736$). At 45° the sine and cosine are equal (0.7071).
What did I learn?	Two acute angles in a right-angled triangle add up to 90° and are called complementary. The side opposite one angle is the side adjacent to the other. So $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$. Wanjiru's 37° at the tank goes with 53° at the top of her line of sight.
How does this explain the phenomenon?	Write in your own words why $\sin 35^\circ = \cos 55^\circ$, using the words opposite, adjacent and hypotenuse, and say how this helps us get closer to finding the height of the Kilimani Secondary School water tank without climbing it.

LESSON 4 (80 minutes (2 x 40-minute lessons)): Choosing the Right Ratio: Solving for Unknown Sides and the First Full Tank-Height Calculation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners choose the correct trigonometric ratio from the known side and the wanted side, set up an equation and solve for an unknown side. They use this to complete the first full tank-height calculation from Wanjiru's data (8 m, 37°).
Knowledge	By the end of the lesson the learner should be able to: determine the sine, cosine and tangent of acute angles in a right-angled triangle and use them to find unknown sides; identify the right-angled triangle in relation to a particular angle (hypotenuse, opposite, adjacent); apply trigonometric ratios to find unknown heights and distances; recognise the use of trigonometry in real-life situations.
Skills	Labelling a triangle from the angle theta; matching known and wanted sides to sin, cos or tan; rearranging an equation to solve for an unknown side; using a scientific calculator in DEG mode; giving trig values to 4 d.p. and final lengths to 1 d.p. with units. NGSS-aligned practices: Developing and Using Models (labelled diagrams), Analyzing and Interpreting Data, Using Mathematics and Computational Thinking, Constructing Explanations.
Attitudes	Core competencies: Self-efficacy (the learner builds confidence by solving real problems with ratios); Learning to learn (the learner learns collaboratively in pairs). Value: Love (the learner shows a caring attitude to others by helping peers check their working). PCI: Self-esteem (the learner gains self-esteem in solving problems that look impossible at first).
Key Inquiry Question	Why is trigonometry important?
Purpose in Storyline	Lessons 1 and 2 gave learners the side names and the constant ratios sin, cos and tan. Lesson 4 turns those ratios into a problem-solving tool. Learners use them to calculate Wanjiru's height for the tank for the first time (8 m, 37° , giving 7.5 m). This is the first answer to the driving question and sets up the clinometer in Lesson 5. Lesson 3 is used only as a check that sin and cos are interchangeable between complementary angles. Inverse ratios (finding an angle) are not taught; all angles are given.
Safety Notes	Learners do no outdoor work in this lesson. Remind them to keep rulers and compasses away from faces and to pack away drawing tools after use. Check that calculators are in DEG mode before starting. Do not let anyone climb the water tank tower or the frame, even when the tank is discussed.

B. LESSON OVERVIEW

Madam Achieng reopens the tank problem: Wanjiru stands 8 m from the tower at Kilimani Secondary School, Nakuru, and reads 37° . Learners predict how they could use the constant ratios from Lesson 2 to turn that into a height. They then work through a 3-step method: (1) label the triangle from the angle theta, (2) pick the ratio that links the KNOWN side and the WANTED side, (3) write the equation, solve and state the answer with units. They practise on Kisumu roof rafter, Kericho stream and Mombasa mango tree problems, and a check on a Kisumu rafter shows sin and cos giving the same answer through complementary angles. Finally they use the same method on Wanjiru's data, adding the 1.5 m eye height to get 7.5 m. The Driving Question Board is updated and the class model gains a 'choose the ratio, solve for the side' step.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Right triangle trigonometry word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners see a sketch of the tank tower with Wanjiru 8 m from the base and the angle 37° marked. They are told that the height of the top of the tank above her eye level is the unknown, x . In pairs they write: (1) which side of the triangle they know, which they want, and which ratio might link them, and (2) a rough guess of x with a reason, remembering that the 35° triangles in Lesson 2 all had $\tan = 0.700$ whatever their size. They record predictions on mini-whiteboards or in their books and hold them up.	Draw the right-angled triangle on the board: Wanjiru at one corner, the tower base at the right angle, the tank top at the third corner. Say: 'Wanjiru knows 8 m and 37° . She never climbed the tower. Which ratio could help her, and why?' Give WAIT TIME of 10-15 seconds in silence. Then cold-call 4 learners, including at least 2 who did not volunteer, and ask each: 'Which side is known, which side is wanted?' Do not confirm or correct. Write all ideas on the board under 'Our predictions'. Say: 'Keep your prediction. We will test it today.'	Prediction and Reasoning from Prior Evidence: learners use Lesson 2's discovery that ratios stay constant to make a testable claim BEFORE any new method is taught. This exposes the idea that the ratio is what links a known side to a wanted side, and gives the class something to confirm or revise later.	Check whether learners (a) label 8 m as adjacent and the tank height as opposite relative to 37° , (b) name $\tan$ or at least give opposite and adjacent as the sides involved, (c) give a reason that uses the idea of constant ratios. Tally on a clipboard how many pairs name $\tan$. Learners who confuse opposite and adjacent are flagged for support during the Observe Phase.
Observe Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Right triangle trigonometry word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners work in pairs through a guided 3-step worked example, then carry out 3 practice tasks with a calculator in DEG mode. Step 1: label the triangle from theta (hypotenuse, opposite, adjacent). Step 2: write down the KNOWN side and the WANTED side and pick the ratio that has both (tan for opposite and adjacent, sin for opposite and hypotenuse, cos for adjacent and hypotenuse). Step 3: write the equation, rearrange and solve. Tasks: (a) Kisumu roof rafter: horizontal span 3.6 m, rafter makes 30° with the horizontal, so rafter = $3.6 / \cos 30 = 3.6 / 0.8660 = 4.2$ m (4.16 m). (b) Kericho stream: a learner on the bank opposite a tree walks 20 m along the bank, the line of sight to the tree makes 40° with the bank, so width = $20 \times \tan 40 = 20 \times 0.8391 = 16.8$ m. (c) Mombasa mango tree: 12 m from the base, elevation 28° , eye height 1.5 m, so height =	Model Task (a) on the board with a think-aloud: 'The span 3.6 m is next to the 30° angle, so it is ADJACENT. The rafter is opposite the right angle, so it is the HYPOTENUSE. Adjacent and hypotenuse, which ratio is that?' Give WAIT TIME of 10-15 seconds. Cold-call 3 learners for the ratio and the reason. Show how to rearrange $\cos 30 = 3.6 / \text{rafter}$ into $\text{rafter} = 3.6 / \cos 30$ and point out that the unknown is on the bottom. Then release pairs to do (b) and (c) alone. Circulate with the prompt: 'Show me which side is known and which is wanted before you touch the calculator.' Check DEG mode on at least 5 calculators. For fast pairs ask: 'Mombasa mango tree: what does the 1.5 m add, and why?' As a check, ask pairs to use the complementary angle 60° with $\sin 60 = 0.8660$ for (a) and compare; this is only a check, since	Structured Problem-Solving Routine (Known-Wanted-Ratio table): the 4-column table makes the decision step visible. Learners see that the ratio is chosen by which two sides are involved, not by guessing. Comparing answers across three different contexts shows that the same reasoning works for rafters, rivers and trees. This links directly to the tank problem.	Look for: (a) correct labelling of the sides before calculation, (b) correct ratio chosen in all 3 tasks, (c) the equation written before the calculator is used, (d) answers within 0.1 of 4.2 m, 16.8 m and 7.9 m, (e) eye height of 1.5 m added in task (c). Note common errors (using sin for adjacent and opposite, multiplying when you should divide, calculator in RAD mode, forgetting the 1.5 m) and tally them on the board for the Explain Phase.

	$12 \times \tan 28 + 1.5 = 12 \times 0.5317 + 1.5 = 6.38 + 1.5 = 7.9$ m. They record each task in a 4-column table: known side, wanted side, ratio chosen, answer.	complementary angles were taught in Lesson 3.		
Explain Phase  VIDEO: Finding reciprocal trig ratios <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Trigonometric ratios in right triangles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the anchoring phenomenon and solve it themselves, step by step, using Wanjiru's data (8 m from the tower, 37° , eye height 1.5 m). In pairs they: (1) draw and label the triangle (right angle at the tower base level with her eye, $\theta = 37^\circ$, adjacent = 8 m, opposite = unknown); (2) name the ratio: opposite and adjacent, so $\tan$; (3) write $\tan 37 = x / 8$, so $x = 8 \times \tan 37 = 8 \times 0.7536 = 6.03$ m above eye level; (4) add the eye height: $6.03 + 1.5 = 7.53$ m, which rounds to 7.5 m. Then they compare their solution with the other tasks and write 2-3 sentences: 'We choose the ratio by ... The tank is 7.5 m high because ...'.	Say: 'Remember this sketch? This morning you predicted. Now you will test it.' Do NOT tell the answer. Give 4 minutes for pairs to work. Circulate and ask: 'Why $8 \times \tan 37$ and not $8 / \tan 37$?' Hold a whole-class discussion. Cold-call 3 pairs, choosing pairs with different working, e.g. one that forgot the 1.5 m and one that added it. Ask: 'Why does the answer 6.03 m not give the height from the ground?' WAIT TIME of 10-15 seconds before each call. Ask: 'Compare with your prediction: what did you get right, and what do you want to change?' Then raise the key idea: 'The ratio $\tan 37 = 0.7536$ depends only on the angle, so Wanjiru can use it at any distance.' Preview, without solving, that Otieno stands 10 m away at 31° , so the class can see in Lesson 8 whether his answer matches.	Claim-Evidence-Reasoning with Prediction Revisit: learners state a claim (the height is 7.5 m), support it with evidence (the calculation) and give reasoning (the ratio depends only on the angle, so it links the 8 m to the height). Revisiting their own predictions turns the lesson into a conceptual change, not only a procedure.	Collect each pair's written solution. Success criteria: labelled diagram, $\tan$ chosen with a reason, correct equation, 6.03 m (accept 6.0 m), eye height added, final answer 7.5 m with units, and a sentence linking to constant ratios. Use a 3-point check: all 6 steps (secure), 4-5 steps (developing), fewer than 4 (needs support). Plan a small support group for the next lesson.
Driving Question Board (DQB) Creation  VIDEO: Intro to the trigonometric ratios <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML:	Learners revisit the class DQB with its three columns, 'Questions', 'Answered' and 'New understandings'. On sticky notes in pairs they (1) move at least one question to 'Answered', for example 'Which ratio do we use to find the height from the distance and the angle?' with the answer 'tan, because the 8 m is adjacent and the height is opposite'; (2) add 'New understandings', for example 'We pick the ratio by the known and the wanted side' and 'We must add 1.5 m eye height'; and (3) add a new question, for example 'How do we measure the angle in the	Say: 'Look at the board. Which question can we now answer, and what is your evidence?' WAIT TIME of 10-15 seconds. Cold-call 3 learners to place a sticky note and justify it. Insist on evidence: 'Which calculation shows that?' Praise good new questions such as the clinometer one and say: 'That is exactly where we are going next lesson.' Do not answer the Otieno question; say: 'We will test that with real data.'	Driving Question Board as a Public Record of Learning: learners see their own questions get answered and new ones appear. This keeps the whole sequence coherent, shows how each lesson moves the work toward the final clinometer measurement, and builds ownership of the investigation.	Check that notes moved to 'Answered' cite a calculation or a diagram as evidence, that the 'New understandings' column includes the idea of choosing the ratio from the known and wanted sides, and that at least 5 new questions are linked to measuring angles or to Otieno's data. Note learners who add no questions.

Sine and Cosine of Complementary Angles (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	first place without a protractor on the tower?' or 'Does Otieno get the same height from 10 m and 31 degrees?'			
Model Building Phase  VIDEO: Trig word problem: complementary angles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Sine and Cosine of Complementary Angles Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners revise their class model of the tank problem. In pairs they redraw their model diagram: the tower, Wanjiru at 8 m, the 37° angle, the labelled sides (adjacent 8 m, opposite unknown, hypotenuse the line of sight), and add a new 'Choose and solve' box with 3 steps: label the sides, pick the ratio from the known and wanted side, write the equation and solve. They add a note in a different colour: 'Height above ground = distance x tan(angle) + 1.5 m'. They then add a line, 'Why it works: the ratio depends only on the angle, not on the size of the triangle'. Finally they add a short flowchart: Known side + wanted side leads to sin, cos or tan.	Say: 'Your model must now show how a number is produced, not only what the sides are called.' Give 6 minutes. Circulate and ask: 'Where does your model show why 8 m and 37 degrees are enough?' Cold-call 3 pairs to show models and ask the class: 'Does this model explain Wanjiru's answer? What is still missing?' WAIT TIME of 10-15 seconds. Draw out the gap: 'How do we actually get 37 degrees from the tower?' Leave the class with the gap for Lesson 5. Optional extension for early finishers: use a complementary angle to check the tank answer using sin and cos, as a Lesson 3 check only.	Cumulative Model Revision: learners add the new 'choose and solve' mechanism to their earlier models, building the explanation of the phenomenon in layers. Marking the changes in a different colour shows how their understanding has grown since Lesson 1 and highlights the missing piece (measuring the angle), which motivates building the clinometer.	Look for models that (a) show correct labelling relative to 37°, (b) include the 3-step choose-and-solve process, (c) include the 1.5 m eye height in the formula, (d) state that the ratio depends on the angle only, and (e) identify a gap about measuring the angle. Use an exit ticket: 'A Kericho tree is 25 m away, elevation 30 degrees. Which ratio would you use to find the height above your eye, and why?' Collect and sort into secure, developing and needs support.

D. TEACHER REFLECTION

After the lesson, review the paired solutions and exit tickets. Which ratio-choice errors were most common (sin vs tan, multiplying vs dividing)? Did learners explain WHY they chose the ratio, or just follow the steps? Did every learner remember to add the 1.5 m eye height? How many learners were secure on the tank calculation, and which learners need extra support before the clinometer in Lesson 5? Was WAIT TIME kept to 10-15 seconds, and did cold-calling include quieter learners? Which new DQB questions will guide Lesson 5, and were calculators in DEG mode throughout?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that to find an unknown side, we label the triangle from the angle, pick the ratio that links the KNOWN side and the WANTED side, write the equation and solve. In the tank problem, Wanjiru is 8 m away at 37 degrees, so $8 \times \tan 37 = 6.03$ m above eye level, and with her 1.5 m eye height the top of the tank is 7.5 m.
What did I learn?	We learned that tan links opposite and adjacent, sin links opposite and hypotenuse, and cos links adjacent and hypotenuse. We learned that the ratio depends only on the angle, so one angle and one side are enough to find another side. We also learned to use $\tan 37 = 0.7536$ in DEG mode and to add the 1.5 m eye height.
How does this explain the phenomenon?	We can now partly explain how the tank height can be found without climbing it: the known distance and the angle give a ratio that lets us calculate the height. We still need to explain how to measure the angle accurately, and to check that Otieno (10 m, 31 degrees) gets the same height, which we will test in later lessons.

LESSON 5 (80 minutes (double lesson of 2 × 40 minutes)): Angle of Elevation and Building the Clinometer

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners define the angle of elevation, build a simple clinometer from a cardboard protractor, straw, string and washer, and read it correctly (angle = degrees between the string and the 90° mark). They test it on the classroom door frame and set up tan for the height, so the water-tank problem moves from calculation on paper to real measurement.
Knowledge	By the end of the lesson the learner should be able to: (1) define the angle of elevation as the angle above the horizontal from the observer's eye up to the object; (2) explain why the protractor reading is 90° minus the angle of elevation, using complementary angles; (3) read a clinometer, where the angle of elevation is the number of degrees between the string and the 90° mark when the straw is sighted on the object; (4) set up height = (distance × tan angle) + 1.5 m for a clinometer measurement.
Skills	Making a simple clinometer using locally available materials; measuring angles of elevation of objects in the immediate environment; reading a scale accurately to the nearest degree; modelling a real situation with a right-angled triangle; setting up tan for a height. NGSS-aligned practices: Developing and Using Models, Planning and Carrying Out Investigations, Constructing Explanations, Engaging in Argument from Evidence.
Attitudes	Love: the learner demonstrates a caring attitude to others when using the clinometer to measure the angle of elevation of objects in the immediate environment (taking turns, steadying a partner's reading). Learners also build self-efficacy and self-esteem by making and using their own instrument, and show responsibility in caring for the materials.
Key Inquiry Question	Why is trigonometry important?
Purpose in Storyline	Lessons 1 to 4 gave learners the right-angled triangle model, the side names, the ratios sin, cos and tan, the complementary-angle link, and a method for solving for an unknown side. Lesson 5 turns the picture of Wanjiru's angle into a real instrument: learners learn what 'angle of elevation' means and build the clinometer they will use outdoors in Lesson 6 and to confirm the tank height in Lesson 8. Today they test it indoors on a target with a known height, the classroom door frame.
Safety Notes	Use sharp scissors and pointed tools carefully. Pass scissors handle-first. Do not sight directly at the sun or bright windows through the straw. Keep the straw away from partners' eyes, and hold it against your own cheek. Washers swing, so do not swing them at classmates. Walk, do not run, while measuring distances in the classroom and keep bags out of the way. Measurements are taken on level ground or a level floor.

B. LESSON OVERVIEW

Madam Achieng reopens the anchoring phenomenon: Wanjiru at Kilimani Secondary School in Nakuru stands 8 m from the tank tower and reads an angle of 37°. But how did she get that angle without climbing? Learners predict how a protractor, a string and a straw could capture an angle of looking upwards, then build a clinometer from a cardboard protractor, a straw taped along the straight edge, and a string with a washer hung from the centre mark. They observe that when the straw is horizontal the string hangs at the 90° mark, and when the straw is tilted up to sight an object the string moves away from 90°. The angle of elevation is the number of degrees between the string and the 90° mark. Learners connect this to complementary angles (the scale reading is 90° minus the angle of elevation). Finally they test the clinometer on the top of the classroom door frame, 2.1 m high, sighted from 3.0 m away with eye height 1.5 m. The expected angle is about 11°. Learners set up height = $(3.0 \times \tan \text{angle}) + 1.5$ m and compare with the known 2.1 m. They then add to the Driving Question Board and revise their cumulative model of how the tank height can be found without climbing.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Solving for a side in right triangles with trigonometry <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Trigonometry Word Problems: Helicopter Angle of Depression Rescue (PLIX) Source: CK-12  Search ARES for similar readings	Learners look at the Lesson 1 diagram of the tank tower with Wanjiru and Otieno. Madam Achieng shows a bare cardboard protractor, a straw, a piece of string and a washer. Individually in their notebooks (3 minutes) they answer: 'Wanjiru read 37° at the tank tower. How could these four items measure the angle at which she looks UP at the top of the tank? Draw your idea and label where the angle is.' They draw a sketch, then pair-share, then a few learners share with the class. Learners also predict: 'What do you think the string will do when the straw points straight ahead, and when it points up at the top of the tank?'	Say: 'Wanjiru never climbed the tank at Kilimani, yet she has an angle of 37° . Today we make the tool that gives us that angle.' Hold up the four items and say: 'Look at these items. Do not touch them yet.' Give 3 minutes of silent drawing, with WAIT TIME of 10-15 seconds after asking 'Where is the angle in your sketch?' before taking any answer. Circulate and note which learners draw the angle from the eye going up (the correct idea), from the ground, or at the top of the tank. Then say: 'Turn to your partner and explain your sketch in 1 minute.' Cold-call 3 pairs with different ideas and write the ideas on the board without judging. Ask: 'Why do you think the string will hang straight down whatever we do?' Take 2 responses. Record the predictions on chart paper to revisit in the Explain phase.	Predict-Observe-Explain with Sketch-and-Share: committing to a drawn prediction makes learners' existing ideas about angle and gravity visible. Later, comparing the sketch with the observed clinometer makes the conceptual change clear and gives learners ownership of the revision.	Look for: (a) sketches that show a horizontal line from the eye and a sight line rising to the top of the tank; (b) learners who note that the string hangs down by gravity; (c) learners who place the angle at the observer's eye and not at the top of the tank. Tally on a checklist how many learners locate the angle correctly. Misconception to note: measuring the angle at the top of the tank or between the sight line and the vertical.
Observe Phase  VIDEO: Measuring angles in degrees <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Angles of Elevation and Depression (Flexbook)	In groups of 4, learners build a clinometer: (1) tape a straw along the straight edge of a cardboard protractor; (2) push a thread through the small hole at the centre mark and knot it; (3) tie a washer to the free end so the string hangs freely over the scale. Learner 1 holds the clinometer and sights a target through the straw; Learner 2 reads the number where the string crosses the scale; Learner 3 records; Learner 4 checks that the straw is steady. Step A (horizontal check): hold the straw level and look at the far wall at eye height; record where the string hangs (the 90° mark). Step	Before building, state safety: 'Scissors handle-first. Straw against your cheek, never pointed at a face. Do not sight at the sun.' Demonstrate the build once while narrating: 'The straw must run exactly along the straight edge. The string must hang from the centre mark.' Use WAIT TIME of 10-15 seconds after asking 'Where does the string hang when the straw is level, and why is it there?' Mark a floor line at 3.0 m from the door with tape and mark the eye-height point on the wall at 1.5 m. Circulate and ask each group: 'Where is the string when the straw is level? What is that number?'	Structured data collection with roles and a class data pool: assigning roles ensures everyone handles the instrument, and pooling all group readings on the board shows learners that real measurements vary slightly but cluster around an expected value. This links directly to the later idea that Wanjiru's and Otieno's readings are to the nearest degree.	Observable indicators: (a) the string hangs at 90° when the straw is level; (b) learners subtract the reading from 90° or read the angle from the 90° mark, not the raw scale number; (c) the straw is aligned along the edge and the string moves freely; (d) the group's angle falls in about 10° to 12° . Use a rating scale (Meeting / Approaching / Not yet) and write the groups' names against each. Note any group that reports 79° as the angle of elevation and prompt it to explain the 90° mark.

Source: CK-12 Search ARES for similar readings	B (tilting up): sight the top of the classroom door frame from 3.0 m away (the eye is at 1.5 m) and record the string position. Learners work out the angle of elevation as the number of degrees between the string and the 90° mark, then compare with the expected value of about 11°. Each learner takes a turn at sighting, reading and recording, and results go in a table: Learner Scale reading Angle between string and 90° mark.	Ask: 'When you tilt the straw up, which way does the string move on the scale?' Prompt groups to record the scale reading before they subtract. If a group gets a value far from 11°, ask: 'Is the straw really on the top of the frame? Is your eye at 1.5 m? Is the string free?' Cold-call 4 groups for their measured angles and write all values on the board; note that they cluster near 11° (some will read 10° to 12°).		
Explain Phase VIDEO: Solving for a side in right triangles with trigonometry <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Angles of Elevation and Depression Practice Source: CK-12 Search ARES for similar readings	Learners use their data to build the explanation. Part 1 (meaning): in groups they complete the definition 'The angle of elevation is the angle ____ the horizontal from the observer's ____ up to the object.' and mark it on a diagram of Wanjiru, the tank and the horizontal line at eye level. Part 2 (complementary angles): they draw the right-angled triangle formed by the protractor's 90° mark, the string and the straw, and explain why the scale reading (for example 79°) and the angle of elevation (11°) are complementary, adding to 90°. Part 3 (tan for height): using the Lesson 4 method, they set up the right-angled triangle for the door frame: the horizontal distance is 3.0 m and the opposite side is the height above eye level. They explain that the door frame is 2.1 m high and the eye is 1.5 m above the floor, so the height above eye level is 0.6 m, and check that $\tan$ of the angle is $0.6 \div 3.0 = 0.2$. They write the general rule: height = (distance $\times$ $\tan$ angle) + 1.5 m. They then use Wanjiru's values (8 m, 37°) to say what the clinometer would show for her reading: string	Ask: 'Where does the angle of elevation begin, at the ground or at the eye? Show me on your diagram.' WAIT TIME of 10-15 seconds. Cold-call 3 learners to place the angle on the board diagram, asking the class for thumbs-up or thumbs-down. Ask: 'Why is the number on the scale not the angle of elevation when the string hangs at 79°?' Take 3 responses and press for the word 'complementary'. Say: 'Remember Lesson 3: complementary angles add to 90°. What is 90° minus 11°?' For Part 3 say: 'This is the same right-angled triangle as in Lesson 4. Which side do we know? Which side do we want? Which ratio connects them?' Cold-call 2 learners to choose $\tan$ and give a reason. Tell them: 'We are not working backwards from 0.2 to an angle; the expected angle is about 11° and we simply check against it.' Ask: 'Why do we add 1.5 m at the end of the height?' Take 2 responses. Ask: 'Wanjiru read 37°. Where would the string hang on the scale?' Cold-call 3 learners. Finally say: 'Go back to your Predict sketch. Circle what you got right and fix what you want	Claim-Evidence-Reasoning with Revisit-the-Prediction: learners state the claim (the angle of elevation is measured from the horizontal at eye level, and the scale reading is its complement), support it with their own clinometer readings, and reason with complementary angles and $\tan$. Connecting Lesson 1 (triangle model), Lesson 3 (complementary angles) and Lesson 4 (setting up $\tan$) shows learners that the new instrument uses ideas they already own.	Exit-slip check (3 minutes, individual): (1) 'Define the angle of elevation in one sentence.' (2) 'The string of a clinometer crosses the scale at 53° when the 90° mark is the level position. What is the angle of elevation?' (Expected: 37°.) (3) 'Write the equation for the height of a target 3.0 m away when the angle of elevation is 11°.' (Expected: height = $3.0 \times \tan 11^\circ + 1.5$ m.) Success criteria: at least 80% define the angle from the horizontal at the eye and use the 90° mark correctly. Note learners who read the scale value directly as the angle or leave out the 1.5 m eye height for reteaching.

	37° away from the 90° mark, scale reading 53°. Finally they revise their Predict sketches.	to change.'		
Driving Question Board (DQB) Creation  VIDEO: Measuring angles in degrees <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Trigonometry Word Problems: Helicopter Angle of Depression Rescue (PLIX) Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board with the columns 'Questions', 'Answered' and 'New understandings'. In groups they write on sticky notes: (1) one question that the clinometer has now answered (for example, 'How did Wanjiru get an angle without climbing?') and move it to 'Answered'; (2) one new understanding (for example, 'The angle of elevation is measured from the horizontal at the eye, and the clinometer reads it between the string and the 90° mark'); (3) one new question that remains (for example, 'How do we measure when the object is below us?' or 'How accurate is a cardboard clinometer outdoors?'). Groups post their notes and the class reads them aloud.	Say: 'Look at the DQB. Which of our earlier questions can we move today, and what is your evidence?' WAIT TIME of 10-15 seconds. Cold-call 4 groups, one per question, to justify moving a note to 'Answered' with evidence from their clinometer. Ask: 'What new question has the clinometer created for you?' Take at least 3 and say that Lesson 6 will take the clinometer outdoors and look at looking down as well as looking up. Ask: 'Which new understanding is most important for finding the height of the tank at Kilimani?' Take 2 responses and have the class vote by show of hands. Tell learners: 'Next lesson, bring your clinometer; we meet the flagpole outside.'	Driving Question Board as a public record of learning: moving questions to 'Answered' with evidence, and adding 'New understandings', shows learners that understanding grows through evidence and that each new tool raises new questions. It keeps the anchoring phenomenon central across lessons.	Check that: (a) each group's 'Answered' note is supported by evidence from today's clinometer data and not just a statement; (b) at least some new questions point to Lesson 6 ideas (looking down, outdoor accuracy, errors); (c) the 'New understandings' notes use the term 'angle of elevation' correctly. Record which groups' notes show misconceptions, such as treating the scale number as the elevation, for the next lesson's starter.
Model Building Phase  VIDEO: Solving for a side in right triangles with trigonometry <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Angles of Elevation and Depression (Flexbook)	Learners revise their cumulative model (a labelled diagram plus a short written explanation) of how the height of the Kilimani Secondary School water tank can be found without climbing. The new parts of the model are: (1) the horizontal line at eye level (1.5 m above the ground) with the angle of elevation marked between it and the line of sight; (2) the clinometer, with the string, the 90° mark and the scale reading; (3) the right-angled triangle with the horizontal distance as the adjacent side and the height above eye level as the opposite side; (4) the equation $\text{height} = (\text{distance} \times \tan \text{angle}) + 1.5 \text{ m}$. They add an annotation:	Say: 'Your model started in Lesson 1 as a triangle with names. Today it gets a real instrument. Add the horizontal at eye level, the angle of elevation, the clinometer and the height rule.' Give 8 minutes of independent work, then 2 minutes of pair review: 'Swap diagrams. Does your partner's angle start at the horizontal at the eye? Is the 1.5 m included?' Circulate and ask: 'Where does the 1.5 m appear in your model?' and 'Which side of your triangle is 6.0 m for Wanjiru's tank?' Cold-call 3 learners to share their sentence, using WAIT TIME of 10-15 seconds before choosing. Close by saying: 'Next lesson we take the clinometer outside to	Cumulative model revision: adding the instrument and the eye-level horizontal to the existing triangle model lets learners see how each lesson's idea fits into one explanation of the phenomenon, and the written 'because' sentence turns the diagram into a causal explanation.	Rubric (Meets expectations if all appear): (a) the angle of elevation is drawn from the horizontal at eye level; (b) the clinometer is labelled with the string and the 90° mark; (c) the scale reading is related to the angle by 90° minus the angle; (d) the equation includes the 1.5 m eye height; (e) the explanation sentence refers to the right-angled triangle and the tan ratio. Mark each model as Meeting, Approaching or Not yet, and record which learners to pair with a peer coach for Lesson 6.

Source: CK-12 Search ARES for similar readings	'Scale reading = 90° – angle of elevation (complementary angles).' Finally each learner writes 2-3 sentences: 'We can find the height without climbing because ____.' and flags one thing they are still unsure of.	Kilimani's flagpole and find out how close we can get to the caretaker's tape.' Collect the models to see who still confuses the scale reading with the angle.		
---	--	--	--	--

D. TEACHER REFLECTION

After the lesson, reflect on: (1) Did learners place the angle of elevation at the observer's eye and measure it from the horizontal, or did some still measure from the vertical or the ground? (2) How many groups first reported the raw scale number (for example 79°) as the angle? What helped them link it to complementary angles from Lesson 3? (3) Did the class pool of readings cluster near the expected 11° , and did learners accept that small differences come from reading to the nearest degree? (4) Were the roles (holder, reader, recorder, checker) shared fairly, so that every learner, including quieter ones, handled the clinometer? (5) Which clinometers failed (string sticking, straw crooked), and what materials fix this before Lesson 6? (6) Which DQB questions are ready for Lesson 6, and what extra practice on tan setups is needed for learners who left out the 1.5 m?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We built a clinometer from a cardboard protractor, a straw, a string and a washer. When the straw is level the string hangs at the 90° mark, and when we sighted the top of the classroom door frame (2.1 m high) from 3.0 m the string moved away from 90° by about 11° .
What did I learn?	The angle of elevation is the angle above the horizontal from the observer's eye up to the object. On the clinometer it is the number of degrees between the string and the 90° mark, so the scale reading is 90° minus the angle of elevation (complementary angles). Height = (distance $\times$ tan angle) + 1.5 m.
How does this explain the phenomenon?	Wanjiru can get an angle for the Kilimani Secondary School water tank without climbing because her clinometer measures the angle of elevation from her eye. That angle, together with her 8 m horizontal distance, gives a right-angled triangle in which tan connects the two, and adding her 1.5 m eye height gives the tank's height.

LESSON 6 (80 minutes (2 x 40-minute lessons)): Angle of Depression and Outdoor Measurement: Measuring the Flagpole at Kilimani Secondary School

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners extend the angle of elevation to the angle of depression, show that the angle of depression from a high point equals the angle of elevation from the lower point, and use the class-made clinometer outdoors to collect and process real field data (the flagpole), explaining why small errors occur. This moves them closer to confirming the water tank height in Lesson 8.
Knowledge	By the end of the lesson, the learner should be able to: (1) define angle of elevation and angle of depression and tell them apart; (2) explain why the angle of depression from a high point equals the angle of elevation from the lower point (alternate angles); (3) use height = (distance x tan angle) + 1.5 m to process clinometer data; (4) identify sources of error and explain why results differ slightly from a tape-measured value.
Skills	Measuring horizontal distance along level ground with a tape; reading a clinometer (degrees between the string and the 90 degree mark); recording data in a table; setting up and solving tan equations; comparing a calculated value with a tape check and describing the error; collaborating in groups of 4. NGSS practices: Planning and Carrying Out Investigations; Analyzing and Interpreting Data; Using Mathematics and Computational Thinking.
Attitudes	Appreciate the use of trigonometry in real-life situations (KICD SLO e); show care and honesty when recording measurements, including readings that do not match expectations; value teamwork and careful use of locally made instruments.
Key Inquiry Question	Why is trigonometry important?
Purpose in Storyline	Lesson 5 gave learners the clinometer and the angle of elevation. Lesson 4 gave them the skill of solving for sides. In Lesson 6 they go outdoors for the first time to measure a real structure (the flagpole, 9.0 m), meet the angle of depression, and see that real measurements carry small errors. This prepares them for Lesson 7 (multi-step problems and error analysis) and for Lesson 8, where clinometer results are reconciled with the water tank at Kilimani Secondary School, Nakuru. Learners should leave knowing a method that gave 9.1 m for a 9.0 m pole can be trusted, and that it is the same method that gave 7.5 m for the tank.
Safety Notes	Keep learners on level, open ground away from roads and drains. Never sight toward the sun. Nobody climbs the flagpole, the tank tower or the veranda rails. Tape measures are not swung or thrown. Washers on strings are held, not swung. Learners stay within sight of Madam Achieng; groups are of 4 with set roles. The veranda task is done from a safe standing position behind the rail, with no leaning over.

B. LESSON OVERVIEW

Madam Achieng opens by reminding the class of the tank puzzle: Wanjiru (8 m, 37 degrees) and Otieno (10 m, 31 degrees) never climbed the tower, yet both got the same height. Learners predict what happens when the observer is HIGH and looks DOWN at a marker on the ground, and whether that angle is related to the angle from the ground looking up. They then make observations in two parts. Part A (veranda): from an eye level 3.0 m above the ground, they sight a ground marker 4.0 m from the veranda base and find the angle of depression is 37 degrees, and check it with $\tan 37 \text{ degrees} = 0.7536$ and $4.0 \times 0.7536 = 3.01 \text{ m}$, about 3.0 m. Part B (field): in groups of 4 they use their clinometers to measure the school flagpole. Sample group Amani stands 15 m from the base, reads 27 degrees, and calculates $15 \times \tan 27 \text{ degrees} + 1.5 = 7.64 + 1.5 = 9.1 \text{ m}$, against Mr. Kamau's tape value of 9.0 m, an error of 0.1 m caused by reading to the nearest degree. In the Explain phase learners use alternate angles to explain depression equals elevation, and analyse why groups' results vary slightly. The DQB and model are updated: the method works outdoors, and answers carry small errors.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Alternate Interior Angles Principles - Basic Source: CK-12  Search ARES for similar videos  HTML: Trigonometry Word Problems: Helicopter Angle of Depression Rescue (PLIX) Source: CK-12  Search ARES for similar readings	Learners recall the tank scenario (Wanjiru at 8 m reading 37 degrees, Otieno at 10 m reading 31 degrees). Madam Achieng sketches a person on the Kilimani veranda looking DOWN at a stone on the ground, then a second person standing at the stone looking UP at the veranda rail. Individually, in notebooks, learners write: (1) a prediction about how the two angles are related (bigger, smaller, equal, or no relationship) with a reason; (2) a prediction of how close the group's flagpole calculation will be to the real height and why it may not be exact. They draw a quick sketch showing where the angle is measured from. They then share with a partner and commit to one prediction on a sticky note for the class chart.	Draw the veranda sketch and say: 'Last lesson you looked UP with your clinometer. Today someone stands high and looks DOWN. Is the angle going to be the same, or different?' Give silent WAIT TIME of 15 seconds before anyone writes. Say: 'Write your prediction and one reason. There is no wrong answer yet.' Circulate and cold-call 4 learners with different predictions (for example one saying equal, one saying bigger, one saying not related). Ask: 'What made you think that?' Record each prediction as a tally on the board without judging. Then ask: 'If our method gave the tank height without climbing it, how close do you think we will get to the flagpole when we check with Mr. Kamau's tape? Exact? Within 1 m? Within 0.1 m?' WAIT TIME 10 seconds. Tell learners we will revisit these predictions in the Explain phase.	Prediction with reasoning (Predict-Observe-Explain): committing to a prediction and a reason in writing, before seeing evidence, exposes learners' initial ideas about angles measured from high and low points. Because the predictions are recorded publicly, learners feel the need for evidence and notice later when the data support or contradict them.	Check that at least 80% of learners wrote a prediction with a reason. Observable indicators: (1) the sketch places the angle at the observer's eye, measured from the horizontal; (2) learners use the words elevation, depression or horizontal; (3) learners give an idea about error or rounding. Note learners who measure the angle from the vertical or from the sloping line to the ground, and plan to address this in the Observe phase. Tally predictions on the board: equal / bigger / smaller / no relation.
Observe Phase  VIDEO: Alternate Interior Angles Principles - Basic Source: CK-12  Search ARES for similar videos  HTML: Angles of Elevation and Depression (Flexbook) Source: CK-12  Search ARES	Part A, veranda (about 20 minutes): groups of 4 meet at the veranda. Madam Achieng marks eye level at 3.0 m above the ground and places a ground marker 4.0 m from the base of the veranda. Learners sight the marker with the clinometer held to the eye, with the straw pointing down at the marker, and observe that the string reads against the protractor, with the angle of depression measured from the horizontal line of sight (the 90 degree mark) down to the line of sight. They find about 37 degrees.	At the veranda, say: 'The straw must point at the marker. Where is the horizontal? Show me with your hand.' Before anyone reads, give WAIT TIME of 10 seconds for learners to locate the 90 degree mark. Cold-call 3 learners: 'Where do you start measuring the angle from?' (from the horizontal, not the ground). Ask: 'What do you notice about the angle when the observer is on the ground looking up at the rail, compared with looking down from the veranda?' WAIT TIME 10 seconds. Outdoors, circulate and prompt: 'Is the tape on level	Structured data collection with role rotation and data table (Planning and Carrying Out Investigations): organising raw readings into a table of distance, angle, tan and height makes patterns and spread visible. Using the veranda case first lets learners see depression and elevation as a pair before they handle noisy outdoor data.	Observable indicators: (1) learners hold the clinometer at eye level and read the angle between the string and the 90 degree mark; (2) tape measured along level ground to the base; (3) recorder completes all table columns with the 1.5 m eye height added; (4) depression reading at the veranda is within about 2 degrees of 37 degrees; (5) calculations give a flagpole height within about 0.5 m of 9.0 m. Use a checklist with 3 levels: independent, with prompts, needs support. Ask 2 groups to show their tables and explain why

for similar readings	They check with $\tan 37^\circ = 0.7536$ and $4.0 \times 0.7536 = 3.01$ m, about 3.0 m. A learner then stands at the marker and sights up to the veranda rail at the eye-level mark, reading an elevation also about 37°. Part B, flagpole (about 40 minutes): each group of 4 gets roles: sighter, string-reader, tape-holder, recorder. They measure a horizontal distance on level ground from the base of the flagpole (Group Amani uses 15 m), sight the top, and record the angle. Group Amani reads 27°. They record data in a table: distance, angle, $\tan$ of angle, distance $\times \tan$, plus 1.5 m eye height, calculated height. Group Amani: $15 \times 0.5095 = 7.64$; $7.64 + 1.5 = 9.1$ m. Each learner in the group takes a turn sighting, and groups repeat the reading and note any differences. Back in class, Mr. Kamau's tape value of 9.0 m is shown for comparison.	ground? Is your 15 m measured to the base, not to the shadow of the pole?' Check that the clinometer is held at eye level and the reader waits for the washer to settle. Say: 'Take the reading twice. If the two readings differ by 1 degree, record both.' Ask each recorder: 'Show me your table. What do you do with the 1.5 m?' Remind: 'Safe distances, no one near the flagpole ropes.' Back in class, show the sample results and Mr. Kamau's 9.0 m, and ask: 'The tape says 9.0 m. We got 9.1 m. Is that a failure?' WAIT TIME 15 seconds.		they added 1.5 m.
Explain Phase  VIDEO: Solving for a side in right triangles with trigonometry <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Angles of Elevation and Depression Practice Source: CK-12  Search ARES for similar readings	Learners complete three sense-making tasks. (1) Explain the equality of angles: using the veranda sketch, they draw the horizontal line through the high eye and the horizontal line through the ground eye, and the line of sight between them. They label the angle of depression (below the upper horizontal) and the angle of elevation (above the lower horizontal) and use alternate angles between parallel lines to explain why both are 37°. They add the right-angled triangle with opposite 3.0 m and adjacent 4.0 m and show that $\tan 37^\circ = 0.7536$ gives $4.0 \times 0.7536 = 3.01$ m, about 3.0 m. (2) Interpret the flagpole result: Group Amani's 9.1 m versus 9.0 m. They	Say: 'Draw two horizontal lines, one at the veranda eye level and one at your eye on the ground. What is true about them?' (parallel). WAIT TIME 15 seconds. Cold-call 3 learners: 'Which angles are equal, and why?' Introduce the term alternate angles if no one uses it. Ask: 'Which would be easier to measure in real life, depression from the roof or elevation from the ground? Why?' Then say: 'Group Amani got 9.1 m and the tape says 9.0 m. Who can tell me what 0.1 m means in a 9 m pole?' Cold-call 2 learners. Challenge: 'If they had read 28° instead of 27°, what would the height be?' Give 10 seconds, then show the calculation: $15 \times 0.5317 = 7.98$, plus 1.5 = 9.5 m.	Claim-Evidence-Reasoning with sensitivity testing: learners claim that depression equals elevation and back it with the veranda measurements and alternate-angle geometry. They also test how a 1 degree change affects the answer, which gives them evidence-based reasoning about error instead of treating 0.1 m as a mistake.	Look for: (1) a diagram with two parallel horizontals, correctly labelled depression and elevation; (2) correct use of alternate angles in the explanation; (3) correct calculation $15 \times 0.5095 + 1.5 = 9.1$ m and the error stated as 0.1 m; (4) at least two valid sources of error, with reading to the nearest degree named as the main one; (5) revised predictions that refer to the data. Common misconception to listen for: measuring the angle of depression from the vertical or from the ground. Use a 3-question exit slip: (a) Define angle of depression. (b) If the depression from a roof is 40°, what is the elevation from the ground? (c) Give one reason why a calculated height may differ slightly from a

	list possible reasons for the 0.1 m error (reading the angle to the nearest degree, string swinging, slight tilt of the straw, tape not exactly straight, level of ground, eye height not exactly 1.5 m) and decide which is the main one. They check how much 1 degree changes the answer: $15 \times \tan 27^\circ = 7.64$ m versus $15 \times \tan 28^\circ = 15 \times 0.5317 = 7.98$ m, which is a change of 0.3 m. (3) Return to the Predict phase and revise their predictions in a different colour of pen. They link the flagpole results to the tank: Wanjiru's and Otieno's 7.5 m, which differ only through rounding.	Ask: 'So what does one degree of reading error do to our answer?' Press for evidence: 'What in the data tells you that?' Avoid giving the answer first. Then ask: 'What does this say about how confident we can be about the tank height of 7.5 m?' WAIT TIME 10 seconds. Ask 4 learners to read out their revised predictions.		tape measurement.
Driving Question Board (DQB) Creation  VIDEO: Alternate Exterior Angles Principles - Basic <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Trigonometry Word Problems: Helicopter Angle of Depression Rescue (PLIX) Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board with three columns: 'Questions', 'Answered' and 'New understandings'. In groups they move earlier questions to 'Answered' where the evidence supports it (for example, 'Can we find a height without climbing?' and 'Does the angle depend on where you stand?'). They write at least one new understanding on a sticky note, such as 'Angle of depression from a high point = angle of elevation from the lower point', 'Field readings give answers that are close, not exact (flagpole 9.1 m vs 9.0 m)', and 'One degree of error changes the flagpole height by about 0.3 m'. They then post new questions, such as 'How do we reduce error?', 'Can we use sin and cos for slanted lengths too, like a rafter or a ramp?', and 'Will the clinometer give 7.5 m for the tank?'.	Say: 'Go to the DQB with your group. Which questions can we now move to Answered? Show me the evidence from today.' Give 2 minutes of silent reading, then WAIT TIME 10 seconds. Cold-call 3 groups: 'Which question are you moving, and what is your evidence?' Ask the class: 'Do we agree?' Then say: 'Each group write ONE new question that today's measuring made you wonder about.' Collect at least 4 questions. Point out to learners that the question about slanted lengths will be explored in the next lesson and that the tank confirmation is coming in Lesson 8. Ask: 'Which new question would help us improve our flagpole result?'	Public question tracking (DQB): moving questions between columns and adding new understandings makes progress visible and keeps the driving question (how to find a height or distance without measuring it directly) alive. Writing questions that arise from error encourages learners to treat uncertainty as something to investigate.	Observable indicators: (1) each group moves at least one question and cites specific evidence from today; (2) new understandings are stated accurately, using the terms depression, elevation and error; (3) new questions are testable and linked to the tank, such as reducing error or using sin and cos; (4) learners listen and respond to other groups' notes. Record which groups wrote questions about error or accuracy to help plan Lesson 7.
Model Building	Learners update their class model of how to find the tank height	Say: 'Your model now needs to explain two new things: what	Cumulative model revision: adding depression, field data and error to	Use a 4-point checklist: (1) diagram has labelled horizontal,

Phase  VIDEO: Alternate Interior Angles Principles - Basic Source: CK-12  Search ARES for similar videos  HTML: Angles of Elevation and Depression (Flexbook) Source: CK-12  Search ARES for similar readings	without climbing it. In groups they redraw the model diagram so it now includes: (a) the tank tower with the observer on level ground, horizontal distance, eye height 1.5 m, angle of elevation; (b) a second sketch with the observer on a high point looking down, showing that depression equals elevation; (c) a short note on the method: height = (distance x tan angle) + 1.5 m, using the Group Amani flagpole as a worked check (15 x tan 27 degrees + 1.5 = 9.1 m, tape 9.0 m); and (d) an 'error box' that lists the sources of error and states that angles are read to the nearest degree. They also write a short caption explaining how the flagpole result makes them more confident in the tank method used by Wanjiru (8 m, 37 degrees) and Otieno (10 m, 31 degrees), who both got 7.5 m.	happens when we look down, and why real measurements are a little off. Add them.' Circulate and ask each group: 'Where in your model does the 1.5 m come in? Where is the horizontal?' WAIT TIME 10 seconds before follow-ups. Cold-call 3 groups to explain their error box. Prompt: 'Is 0.1 m in 9.0 m a big error? How could we check?' (about 1%). Encourage each group to add one limit: 'Where might your model not work?' (for example, sloping ground, a tape not level). Close by asking, 'What do you expect to happen when we compare our clinometer results with Mr. Kamau's tape value for the tank?' and tell them that comparison happens in Lesson 8. Collect model drawings for a gallery walk next lesson.	the existing tank model shows how scientific and mathematical models grow with new evidence. Stating limits and error ranges helps learners see a model as a tool whose accuracy can be judged.	line of sight, angle and 1.5 m eye height; (2) depression = elevation is shown using alternate angles; (3) method formula is correct and the flagpole check (9.1 m vs 9.0 m) is cited; (4) the error box names at least two sources and states that rounding to the nearest degree is the main one. Note groups that include a limit of the model. Use this evidence to plan Lesson 7 support for learners who confuse the angle's reference line.
---	--	---	--	---

D. TEACHER REFLECTION

Before the lesson: Did you check that the ground by the flagpole is level and that you have at least one tape measure of 15 m or more per two groups? Have the clinometers been checked so that the string hangs freely and the straw is firmly taped? During the lesson: Which learners found the 90 degree mark and horizontal difficult? Did learners treat the 0.1 m difference as a failure or as expected error? Did anyone ask whether the equality of depression and elevation works for any angle (carry this to the DQB)? After the lesson: Review the exit slips. Which learners can state the definition of depression and apply alternate angles? Which groups got heights more than 0.5 m from 9.0 m, and do they need to repeat the reading? Plan for Lesson 7 which will use sin, cos and tan for multi-step slant-length problems, and for Lesson 8 where learners will reconcile their clinometer data with the tank at Kilimani Secondary School.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe? Record the veranda results (eye level 3.0 m, marker 4.0 m from the base, angle of depression about 37 degrees, with $4.0 \times \tan 37 \text{ degrees} = 3.01 \text{ m}$, about 3.0 m) and our flagpole data (distance, clinometer angle, calculated height, tape value 9.0 m).
What did I learn?	What did we learn? Write what the angle of depression is, how it relates to the angle of elevation from the lower point (alternate angles), and how field data compare with the tape measurement (for Group Amani: 15 m, 27 degrees, 9.1 m versus 9.0 m).
How does this explain the phenomenon?	How does this help explain the phenomenon? Explain, in your own words, how the height of the water tank can be found without climbing it, why the answers are close to the real height but not exact, and what causes the small error when we read angles to the

	nearest degree.
--	-----------------

LESSON 7 (80 minutes (2 x 40-minute lessons)): Multi-Step Problems and Error Analysis: Which Ratio Do I Choose?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners choose sin, cos or tan in unfamiliar multi-step elevation and depression contexts (slant lengths and vertical heights), spot and correct typical errors, and evaluate their Lesson 1 predictions about the Kilimanjaro water tank.
Knowledge	By the end of the sub strand, the learner should be able to: b) determine the sine, cosine and tangent of acute angles in a right-angled triangle; d) apply trigonometric ratios to angles of elevation and depression. In this lesson: the choice of ratio depends on which side is known and which is wanted (opposite, adjacent, hypotenuse relative to the angle). Slant lengths often need sin or cos, while vertical heights from a horizontal distance need tan. Eye height of 1.5 m is added only when the sighting is made from eye level.
Skills	Identify the right-angled triangle in a word problem and draw a labelled sketch; name opposite, adjacent and hypotenuse from the given angle; choose sin, cos or tan; set up and solve the equation; check DEG mode; state answers with units and appropriate rounding; critique worked solutions to find errors (wrong ratio, calculator in RAD mode, forgetting the 1.5 m eye height, mixing up the angle); evaluate earlier predictions against evidence. NGSS-aligned practices: Analyzing and Interpreting Data, Using Mathematics and Computational Thinking, Engaging in Argument from Evidence, Constructing Explanations.
Attitudes	Core competency: Self-efficacy: the learner makes observations and establishes relationships between the ratios. Learning to learn: the learner learns collaboratively when checking each other's work. Value: Love: the learner shows a caring attitude to others when giving feedback on mistakes. PCI: Self-esteem: the learner treats errors as steps in learning.
Key Inquiry Question	Why is trigonometry important? (KICD key inquiry question)
Purpose in Storyline	Lessons 4, 5 and 6 taught learners to set up equations for unknown sides, to read the clinometer and to understand elevation, depression and field errors. Lesson 7 stretches those skills to unfamiliar contexts, checks that learners can choose the right ratio, and uses Lesson 1 predictions to show how much the class has learned. It prepares learners for Lesson 8, where they write the full tank solution and reconcile it with Mr. Kamau's tape check.
Safety Notes	This lesson is indoors and desk-based. No climbing is needed. Learners should keep walkways clear and not stand on desks or chairs to sight objects. Calculators should be handled with care, and learners should share them fairly.

B. LESSON OVERVIEW

Madam Achieng opens with the question: 'Wanjiru and Otieno found the tank height using tan. Would tan work for every problem?' Learners first recall their Lesson 1 predictions about how to find the tank height without climbing. They then attempt three unfamiliar problems: a Rift Valley farmer sighting a cliff (tan), a Watamu diver's weighted line (sin), and a Nairobi ramp (sin). In the Observe phase they collect evidence by solving and sorting problems into 'known side / wanted side' ratio choices. In the Explain phase they critique four worked solutions containing typical errors, including a calculator in the wrong mode and a forgotten eye height, and build a ratio-choice decision guide. They then revisit Lesson 1 predictions, add to the Driving Question Board (DQB), and revise the class model so that it includes how to choose the ratio and how to detect errors. This links directly to the phenomenon: Wanjiru at 8 m and 37° and Otieno at 10 m and 31° both obtain 7.5 m using height = distance $\times$ $\tan(\text{angle}) + 1.5$ m, and the same decision skills will be needed in Lesson 8.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Trigonometry Word Problems Principles - Basic <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Right triangle trigonometry word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners take out their Lesson 1 prediction sheets about how to find the height of the Kilimanjaro Secondary School water tank without climbing it. Individually (3 minutes), they reread and mark each idea with a tick (still holds), a question mark (unsure) or a cross (needs revising). Then they see this prompt on the board: 'A diver at Watamu lets out a weighted line 12 m long and it makes 50 degrees with the horizontal water surface. We know the angle and the line length (the slanted side). Which ratio will give the depth, sin, cos or tan? Why?' Learners write a prediction with a short reason and sketch a quick triangle. They then pair up and compare their choices.	Say: 'Take out your Lesson 1 predictions. Read them silently and mark each one: tick, question mark or cross.' Give WAIT TIME of 10-15 seconds before circulating. Draw the Watamu scenario on the board without labelling sides and say: 'Before we calculate anything, predict: which ratio, and why? Write your reason in one sentence.' Give 10-15 seconds of WAIT TIME. Cold-call 3 learners for different predictions and record every prediction on the board without judging. Ask: 'Does anyone have a different idea? What in the problem makes you choose that ratio?' Remind learners: 'Wanjiru's tank used tan. Is this the same type of problem?' Do not confirm yet; say: 'We will test these predictions with evidence today.'	Prediction with justification (Predict-Observe-Explain). Writing a reason before calculation exposes the ideas learners hold about ratio choice, such as always using tan because the tank used tan. Revisiting Lesson 1 predictions shows how their thinking has changed since the start of the unit.	Check prediction sheets for (a) whether learners name the known and wanted sides in their reasoning, (b) whether any learner insists that tan is always the answer, (c) whether learners can sketch the right-angled triangle correctly. Tally on the board how many chose sin, cos or tan. Note learners who cannot start a sketch for follow-up support.
Observe Phase  VIDEO: Trigonometry Word Problems Principles - Basic <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Right triangle trigonometry word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners solve three problems drawn from Kenyan contexts. They draw a labelled triangle for each, mark the angle theta, label hypotenuse, opposite and adjacent, and choose and justify a ratio. (a) A Rift Valley farmer stands 150 m from the base of a cliff and measures an angle of elevation of 40 degrees (eye height ignored). Find the cliff height. (b) A Watamu diver's weighted line is 12 m long and makes 50 degrees with the horizontal surface. Find the depth. (c) A Nairobi ramp is 5 m long along the slope and rises at 8 degrees to the horizontal. Find the vertical rise. They confirm calculators are in DEG mode, calculate, and record in a table: Problem Known side Wanted side Ratio chosen Working	Distribute the problem sheet and ruled tables. Say: 'For each problem, draw the triangle first. Label the three sides from the angle before you pick a ratio.' Circulate and ask: 'Which side do you know? Which side do you want? Where is the right angle?' Give WAIT TIME of 10-15 seconds after each question before giving any hint. Check that calculators show DEG. If a group says 'I used tan for the ramp', ask: 'Is the 5 m the opposite, adjacent or hypotenuse side?' Do not hand over the answer. After 15 minutes, cold-call 3 groups, one per problem, to share their ratio choice and answer. Ask the class: 'Is the answer reasonable? Could the ramp rise be more than 5 m?' Record answers on the board and the ratio choices in a class table.	Data table and pattern-seeking. Recording known side, wanted side and ratio for several problems lets learners see a pattern: opposite and adjacent give tan, opposite and hypotenuse give sin, adjacent and hypotenuse give cos. The reasonableness check builds number sense and ties to the sources of error from Lesson 6.	Look for: correctly labelled triangles; ratio choice supported by naming the two sides; answers of about 126 m, 9.2 m and 0.7 m within rounding; DEG mode checked; units written. Use a checklist during circulation. Learners who label the hypotenuse incorrectly or choose a ratio without naming sides need a re-teach of side naming from Lesson 1 using a small sketch.

	Answer. Expected results: (a) $150 \times \tan 40 = 150 \times 0.8391 = 125.9$ m, about 126 m; (b) $12 \times \sin 50 = 12 \times 0.7660 = 9.2$ m; (c) $5 \times \sin 8 = 5 \times 0.1392 = 0.70$ m (0.7 m). They then compare answers with another group and resolve any disagreements, finally checking whether each answer is reasonable (for example, the ramp rise cannot be larger than the 5 m slope length, and the depth cannot exceed 12 m).	Ask: 'What pattern do you notice between the sides we know and the ratio we picked?'		
Explain Phase  VIDEO: Trigonometry Word Problems: Principles - Basic <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Trigonometry Word Problems: Helicopter Angle of Depression Rescue (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners now critique four worked solutions written by 'fictional classmates' and find the error in each. Error 1: In the cliff problem, 'Mwende' used $150 \times \sin 40 = 96.4$ m (wrong ratio: she had opposite and adjacent, so tan was needed). Error 2: 'Baraka' calculated $12 \times \sin 50$ and got 0.0 (approximately 0.8) because his calculator was in RAD mode (wrong mode). Error 3: 'Zawadi' calculated the Kilimanjaro tank height as $8 \times \tan 37 = 6.03$ m and said this was the tank height, forgetting to add the 1.5 m eye height (the answer should be 7.5 m). Error 4: 'Kiprop' used the angle 8 degrees in the ramp problem but put the angle at the wrong end, using $\cos 8$ and getting 4.95 m (wrong ratio or wrong angle position). Groups write a corrected solution and a one-sentence tip for avoiding that error. The class then builds a Ratio Choice Decision Guide poster: (1) Draw and label the triangle. (2) Name the known side and the wanted side. (3) Choose the ratio linking them. (4) Check DEG mode. (5) Add the 1.5 m eye height if the angle was measured from eye level. (6) Check that the answer is reasonable. Finally they explain, in	Say: 'Four students at another school made mistakes. Your job is to be the detective. Find the error, fix it and write a tip.' Give 10-15 seconds of WAIT TIME after posing each error. Cold-call 4 learners, one per error, to explain what went wrong. Probe: 'How could you have caught that without knowing the right answer?' For Error 3, ask: 'Where does Wanjiru's eye sit above the ground? Where is the top of the tank measured from?' Ask: 'Is it a problem to make mistakes?' and affirm: 'Finding errors is how mathematicians improve.' Guide the class to create the decision guide; write each step as learners offer it, and ask 2 learners to challenge any unclear step. Use the question: 'Wanjiru stands 8 m away and Otieno 10 m away. Why do their angles differ but the height does not?' and allow 10-15 seconds of WAIT TIME before taking 3 responses.	Error analysis and consensus model building. Diagnosing other people's mistakes lets learners recognise their own typical errors without embarrassment. Building a shared decision guide turns separate fixes into a transferable procedure, and explaining why both learners obtain 7.5 m links the procedure to the underlying idea that the ratio depends on the angle only.	Listen for learners who (a) correctly identify wrong ratio, RAD mode, missing eye height and misplaced angle; (b) state the corrected ratio with sides named; (c) mention that the 1.5 m is added only when the angle is measured from eye level; (d) explain the two-learner consistency using the angle-ratio idea. Use a thumbs-up self-check: 'How confident are you in choosing the ratio?' (up, sideways, down) and note learners who need additional practice.

	two sentences, why Wanjiru and Otieno both get 7.5 m even though their angles differ (the ratio opposite divided by adjacent depends only on the angle, so the closer learner has a larger angle, and the calculation compensates).			
Driving Question Board (DQB) Creation  VIDEO: Trigonometry Word Problems Principles - Basic <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Angles of Elevation and Depression (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners return to the DQB, which has the columns 'Questions', 'Answered' and 'New understandings'. In groups they read the questions posted since Lesson 1 and decide which ones can now be moved to 'Answered', such as 'Which side do we use?', 'Does the size of the triangle matter?' and 'How do we know which ratio to use?'. They write evidence from today (for example, the Watamu, cliff and ramp problems) on sticky notes. They then write new understandings, such as 'The ratio depends on the known side and the wanted side' and 'Always check DEG mode and eye height'. Finally each group adds one new question for Lesson 8, for example 'How close will our clinometer answer be to Mr. Kamau's measured value?' or 'How can we show the full solution clearly?'	Say: 'Look at our DQB. Which questions can we answer now? Show me the evidence from today.' Give 10-15 seconds of WAIT TIME before groups move any note. Cold-call 3 groups to justify why a question has moved to 'Answered' and ask the class: 'Do you agree? What evidence supports that?' If a group cannot give evidence, keep the note in 'Questions'. Prompt: 'What new question do we have now that we did not have in Lesson 1?' Ask each group to post exactly one new question. Highlight a question about accuracy and say: 'Lesson 8 will help us test this when we compare with Mr. Kamau's tape check.'	Driving Question Board as a public record of progress. Moving questions to 'Answered' only with evidence encourages claim-evidence reasoning, and new questions show how understanding has deepened and keep the driving question alive.	Check that questions moved to 'Answered' are supported by evidence from the lesson, that 'New understandings' use correct language (opposite, adjacent, hypotenuse, sin, cos, tan), and that new questions point to accuracy and complete solutions. Observe whether all group members contribute.
Model Building Phase  VIDEO: Trigonometry Word Problems Principles - Basic <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos	Learners revisit the class model for the phenomenon 'How tall is the water tank without climbing it?' and the Lesson 1 predictions. Each learner writes an Evaluation of Predictions note: what my original prediction said, what evidence supports or contradicts it, and how I would change it. They then add to the group model: a labelled diagram of the Kilimani tank tower with Wanjiru at 8 m and 37 degrees and Otieno at 10 m and 31 degrees, the 1.5 m eye height,	Say: 'Open your Lesson 1 prediction. Using today's evidence, how would you change it? Write what stays, what changes and why.' Give 10-15 seconds of WAIT TIME before learners start writing. Circulate and ask: 'Which part of your first prediction was closest to the method Wanjiru and Otieno used?' For the model, ask: 'Where does 1.5 m appear in your diagram? Why is it there?' Cold-call 3 learners for their exit task ratio choice. For the gallery walk,	Cumulative model revision with prediction evaluation. Comparing Lesson 1 predictions with current understanding shows learners their conceptual change and strengthens metacognition. Adding the error box makes the model more complete and usable.	Look for evaluation notes that cite evidence (for example, ratios depend on angle, tan links opposite and adjacent), models that include correct labelling and the 1.5 m eye height, and exit task answers that choose tan with correct sides. Use a simple 3-level rubric: complete (all four elements), developing (two or three), beginning (one or none). Learners at 'beginning' get targeted support in Lesson 8.

 EXERCISE: Angles of Elevation and Depression Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	the equation height = distance x $\tan(\text{angle}) + 1.5$ m, and the decision guide steps. They also add a 'Watch out' box listing four typical errors (wrong ratio, RAD mode, no eye height, wrong angle position). Learners complete a short exit task: a surveyor on the Thika Superhighway stands 20 m from the base of a pylon and measures an elevation of 35 degrees; decide the ratio and write the first line of the equation only ($\tan 35 = h$ divided by 20). Groups share their updated models in a gallery walk and leave one comment on another group's model.	say: 'Leave one comment that begins with: I noticed that...' Close by saying: 'In Lesson 8 you will write the full solution and compare it with Mr. Kamau's tape check of the tank.' Do not reveal the tape value.		
--	---	---	--	--

D. TEACHER REFLECTION

After the lesson, reflect: (1) How many learners correctly named the known and wanted sides before choosing a ratio? (2) Which error was most common in the critique task, and what does that say about earlier lessons? (3) Did learners hold the idea that tan is always used because of the tank problem, and did the Watamu and ramp problems change it? (4) Did learners connect Lesson 1 predictions to evidence with specific reasons? (5) Which learners were over-reliant on peers during the group tasks, and how can they be supported before Lesson 8? (6) Were calculators shared fairly and were all in DEG mode? Plan extra practice or a short revision of side naming for learners who still struggle.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe? Record the three problems (Rift Valley cliff, Watamu diver, Nairobi ramp): the known side, the wanted side, the ratio we chose, and our answers (about 126 m, 9.2 m and 0.7 m). Add the four errors found in the critique task.
What did I learn?	What did we learn? Write how to choose sin, cos or tan from the known and wanted sides, when to add the 1.5 m eye height, why DEG mode matters, and how our Lesson 1 predictions have changed.
How does this explain the phenomenon?	How does this help explain the phenomenon? Explain in your own words why Wanjiru (8 m, 37 degrees) and Otieno (10 m, 31 degrees) both get a tank height of 7.5 m, and how choosing the correct ratio and avoiding common errors lets us find the height of the water tank without climbing it.

LESSON 8 (80 minutes (2 x 40-minute lessons)): Solving the Water Tank Problem: Final Explanation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To bring together everything learned in Lessons 1-7 so that learners can write a complete labelled diagram and written solution for the water-tank problem, reconcile their own clinometer results with Mr. Kamau's tape-measured height, and give a final explanation of how trigonometry lets us find heights and distances without measuring them directly.
Knowledge	By the end of the lesson the learner should be able to: (a) apply trigonometric ratios to angles of elevation and depression; (b) identify the right-angled triangle in relation to a particular angle (opposite, adjacent, hypotenuse); (c) determine the sine, cosine and tangent of acute angles in a right-angled triangle; (d) recognise the use of trigonometry in real-life situations. Learners will explain that $\text{opposite} \div \text{adjacent}$ depends only on the angle, not the size of the triangle, which is why Wanjiru (8 m, 37°) and Otieno (10 m, 31°) both get 7.5 m.
Skills	Constructing a complete labelled diagram; selecting a ratio from the known and wanted sides; setting up and solving $\text{height} = (\text{distance} \times \tan \text{angle}) + 1.5 \text{ m}$; stating answers to 1 d.p. with units; evaluating accuracy and rounding error; constructing an evidence-based written explanation and revising a model (NGSS: constructing explanations, developing and using models, engaging in argument from evidence, analysing and interpreting data).
Attitudes	Self-efficacy as learners see they can find a height they cannot climb; love and care for peers when they use the clinometer together and give kind, specific feedback; appreciation of trigonometry in surveying, engineering and forestry; self-esteem from presenting a final solution. KICD core competencies: self-efficacy and learning to learn. Value: love. PCI: self-esteem.
Key Inquiry Question	Why is trigonometry important?
Purpose in Storyline	This is the culmination of the storyline. In Lesson 1 learners could only predict how to find the tank height. They then derived sin, cos and tan (L2), related complementary angles (L3), solved for unknown sides (L4), built and read a clinometer (L5), collected field data and met errors (L6), and solved multi-step problems (L7). In Lesson 8 they use all of this to write the final explanation, reveal Mr. Kamau's tape value of 7.5 m, and close the explanatory loop.
Safety Notes	Any outdoor re-check of the flagpole or tank data stays on level ground, away from the tank tower base, steel frame and any ladder. Learners never climb the tower or frame. Clinometers are used only for sighting the object: learners must never sight at the sun. Move in pairs, keep within the teacher's sight and keep tape measures and strings from tangling at feet.

B. LESSON OVERVIEW

Madam Achieng opens with the question learners could not answer in Lesson 1: how tall is the water tank at Kilimani Secondary School, Nakuru, if nobody can climb it? Learners first make a final prediction of the tank height from Wanjiru's and Otieno's data and compare it with their Lesson 1 predictions. They then write a complete solution (diagram, triangle, ratio, calculation, units, explanation) for Wanjiru (8 m, 37°) and Otieno (10 m, 31°), checking that both give 7.5 m. Mr. Kamau's tape check of 7.5 m is revealed. Learners use the Group Amani flagpole example (9.1 m against 9.0 m) to discuss why small differences arise from reading angles to the nearest degree. They then complete the Driving Question Board, compare their Lesson 1 model with a final model, and write the Final Explanation answering the driving question and the key inquiry question: Why is trigonometry important?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Right triangle trigonometry word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners open their Lesson 1 notebook pages and re-read their first prediction for the tank height. Individually (3 minutes) they write a new, final prediction on a sticky note: 'Wanjiru stands 8 m from the tower and reads 37°. Her eye is 1.5 m above the ground. I predict the top of the tank is ____ m high because ____.' They also predict whether Otieno (10 m, 31°) will get the same height as Wanjiru, and why a smaller angle at a greater distance could still give the same answer. They place the notes in a 'Prediction wall' next to their Lesson 1 notes.	Say: 'Last time, we could only guess. Today, you have all the tools. Before we calculate, I want your best prediction, and I want a reason.' Draw the tower, Wanjiru and Otieno on the board with distances and angles only, not the answer. Ask: 'Who will get the larger angle, Wanjiru or Otieno? Why?' WAIT TIME 10-15 seconds before taking any answer. Cold-call 4 learners to share their prediction and reason. Ask 2 of them: 'What changed from your Lesson 1 prediction?' Do not say whether any prediction is right. Record the range of predictions on the board.	Prediction revisited (Predict-Observe-Explain). Learners compare their present thinking with their Lesson 1 thinking, making their own progress visible and giving them a stake in finding out whether their reasoning holds. Naming the reason behind each prediction exposes whether learners rely on ratio ideas or on guessing.	Listen for predictions that mention tan, opposite and adjacent, or the idea that a closer observer sees a larger angle. Look for numbers in a sensible range (above 1.5 m and below about 15 m). Note learners who still say they must measure the tank directly or climb it. Tick on a class list who gives a ratio-based reason versus a guess.
Observe Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Right triangle trigonometry word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners gather evidence from the full anchor data. (1) In pairs they read Wanjiru's and Otieno's data cards: Wanjiru 8 m, 37°; Otieno 10 m, 31°; eye height 1.5 m for each. (2) Each pair draws a large labelled diagram for each learner on a plain page or graph paper: tower, level ground, observer, 1.5 m eye height, horizontal line at eye level, angle θ at the eye, and the sides labelled opposite, adjacent and hypotenuse. (3) They compute: $\tan 37^\circ = 0.7536$; $8 \times 0.7536 = 6.03$ m above eye level; $6.03 + 1.5 = 7.53$ m, about 7.5 m. Then $\tan 31^\circ = 0.6009$; $10 \times 0.6009 = 6.01$ m; $6.01 + 1.5 = 7.51$ m, about 7.5 m. Calculators are in DEG mode. (4) Pairs swap and peer-check using a checklist: diagram labelled? correct triangle? tan chosen and why? units? eye height added? (5) Mr. Kamau's tape result is revealed: the top of the tank is 7.5 m. (6) Group	Hand out data cards and say: 'Draw first, calculate second. If I cannot see your triangle, I cannot see your thinking.' Circulate and ask: 'Which side is opposite your angle? Which side do you know? Why tan and not sin?' WAIT TIME 10-15 seconds after each question. Check calculators are in DEG mode: ask 3 pairs to show you $\tan 37^\circ = 0.7536$. Watch for common slips: forgetting the 1.5 m eye height, using 7.53 as the distance, or using sin instead of tan. Prompt: 'Is 6.03 m the height of the tank or the height above your eye?' After peer checking, hold the tape result back for 10 seconds of silence, then say: 'Mr. Kamau climbed the tower during maintenance and measured with a tape. He got 7.5 metres.' Ask 3 cold-called learners: 'How does 7.53 compare with 7.5? How does 7.51 compare?' Then show the flagpole example: 'Group Amani	Evidence-based consistency check and structured peer review. Two different measurements (different distance, different angle) converging on one height, confirmed by an independent tape measurement, gives learners direct evidence that the method works. The checklist makes the five solution steps explicit so learners internalise the structure of a complete written solution.	Check diagrams: eye-level horizontal drawn, 1.5 m shown separately from 6.0 m, angle at the eye (not at the tank), sides correctly labelled. Check working: tan chosen with reasoning, tan values to 4 d.p., $8 \times 0.7536 = 6.03$, $+ 1.5 = 7.53$, rounded to 7.5 m with units. Ask pairs to state in one sentence why Otieno and Wanjiru agree. Record how many pairs correctly add 1.5 m on the first attempt and revisit those who do not.

	Amani's flagpole data (15 m, 27°, $15 \times 0.5095 + 1.5 = 9.1$ m against the caretaker's 9.0 m) is displayed for comparison.	calculated 9.1 m. Mr. Kamau's tape said 9.0 m. Where could the 0.1 m come from?		
Explain Phase  VIDEO: Trigonometry Word Problems Principles - Basic <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Trigonometric ratios in right triangles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners work in groups of 4 to write a Complete Written Solution for the water tank on a poster or in their books, following the five steps: (1) Labelled diagram; (2) Identify the triangle and the known and wanted sides relative to θ ; (3) Choose the ratio and give the reason (opposite and adjacent, so tan); (4) Calculate: height = (distance $\times$ tan angle) + 1.5 m, with units; (5) State the height as a sentence and explain why Wanjiru and Otieno agree. They then discuss the accuracy question: why 7.53 m and 7.51 m are not exactly 7.50 m and why Group Amani got 9.1 m instead of 9.0 m. They link this to angles read to the nearest degree. For example, learners test $15 \times \tan 26^\circ + 1.5$ and $15 \times \tan 28^\circ + 1.5$ using the calculator to see how a 1° change shifts the answer. Groups also explain in two or three sentences why the triangle size does not matter, using Lesson 2's idea that opposite $\div$ adjacent depends only on the angle. Finally, groups briefly state why sin and cos would not be suitable for this problem (we know neither hypotenuse nor have a use for it), and that $\sin 53^\circ$ would equal $\cos 37^\circ$ as complementary angles. Each group presents to another group, and that group gives one 'glow' and one 'grow'.	Say: 'Imagine a Form One learner in Nakuru who missed our lessons. Your solution must let that learner find the tank height alone.' Display the five-step frame. Circulate and use probing questions: 'Why does your answer need the 1.5 m?' 'What does $\tan 37^\circ = 0.7536$ mean in words?' (For every 1 m along the ground, the line of sight rises 0.7536 m.) 'Why is the triangle at Otieno's position bigger but the same height answer?' WAIT TIME 10-15 seconds. Cold-call 3 groups to explain the rounding error and take a total of 3 different explanations. Then run the 1° sensitivity test: ask half the class to compute $15 \times \tan 26^\circ + 1.5$ (8.3 m) and half $15 \times \tan 28^\circ + 1.5$ (9.5 m); with $15 \times \tan 27^\circ + 1.5 = 9.1$ m, ask 'What does a 1° slip do over 15 m?' Link: 'A flagpole 9.0 m is read at 9.1 m. Is the method wrong, or is the reading limited?' Press for the claim-evidence-reasoning structure: 'Claim: _____. Evidence: _____. Reasoning: _____.' Pose the connecting question: 'Surveyors on the Thika Superhighway and foresters in the Mau Forest use this idea. What stays the same, and what changes?' Cold-call 2 learners.	Claim-Evidence-Reasoning (CER) with a structured written solution frame. Learners must connect a claim (the tank is 7.5 m) to evidence (two sets of clinometer data and the tape check) and to the reasoning (the ratio depends only on the angle). The sensitivity test builds quantitative reasoning about measurement error, and gallery-style peer feedback improves clarity of explanations.	Use a 4-point checklist for each group's solution: (a) labelled diagram, (b) triangle and ratio justified, (c) calculation with 1.5 m added and units, (d) explanation that the ratio depends on angle and not triangle size. Look for explanations that mention the nearest-degree reading limit, and for correct vocabulary (opposite, adjacent, elevation, tangent). Note learners who cannot explain why both learners get the same height, and re-teach through the Lesson 2 35° triangle table if needed.
Driving Question Board (DQB) Creation  VIDEO:	The class returns to the Driving Question Board from Lesson 1, which has the columns 'Questions', 'Answered' and 'New understandings'. In groups,	Say: 'Our board started with wonderings. Let us see which ones we have earned the right to answer.' Read each question aloud and WAIT TIME 10-15	Driving Question Board as a record of conceptual change. Moving questions to 'Answered' with evidence makes the learning trail visible and links lessons	Count how many questions each group moves with a specific evidence reference to a lesson. Check 'New understandings' for accuracy: ratio depends on angle

Intro to the trigonometric ratios <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Trigonometry Word Problems: Helicopter Angle of Depression Rescue (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	learners reread every question on the board and decide which can now be moved to 'Answered' (for example: 'How can we find the height without climbing?', 'What do sin, cos and tan mean?', 'Why do two learners get different angles?', 'How do we build a clinometer?'). For each moved question, they write the answer and a piece of evidence (such as the Lesson 2 table, the clinometer reading, or the Group Amani data) on a sticky note. They add 'New understandings' notes, such as 'The ratio opposite ÷ adjacent depends only on the angle', 'We must add eye height', and 'Reading to the nearest degree causes small errors'. They also add any remaining or new questions, such as 'How could we find an angle if we know two sides?', or 'What about triangles without a right angle?', which point to future learning.	seconds before groups decide. Ask: 'What evidence lets us move this one?' Cold-call 5 learners from different groups, each to defend moving one question using a specific lesson (for example: 'Lesson 5, our clinometer reading'). Challenge any answer without evidence: 'Can you show me where we saw that?' Identify any question not fully answered and treat it honestly: for example, 'Finding an angle from two sides is for a later lesson,' and mark it as a future question. Ask the final DQB question: 'Which question surprised you most by how well we can now answer it?'	together. Keeping unresolved questions respects authentic inquiry and shows that science and mathematics continue to generate new questions.	only, add 1.5 m, nearest-degree error. Look for learners who cite evidence versus those who simply assert. Note which questions remain unresolved and whether they are reasonable extensions, not misunderstandings.
Model Building Phase  VIDEO: Solving for a side in right triangles with trigonometry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Right triangle trigonometry word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Individually, learners take out their Lesson 1 model, drawn before any instruction (usually a drawing of the tower and learners with a guess). They create a Final Model on a new page: a labelled diagram of the tank tower with Wanjiru at 8 m, 37°, Otieno at 10 m, 31°, the 1.5 m eye height, the horizontal at eye level, the right-angled triangle, the sides labelled opposite, adjacent and hypotenuse, and the equation height = (distance x tan angle) + 1.5 m. They add annotations: $\tan 37^\circ = 0.7536$ and $\tan 31^\circ = 0.6009$, both giving 7.5 m, and Mr. Kamau's tape check of 7.5 m. They then write their Final Explanation (6-8 sentences) answering: 'How can we use the angles and one side of a right-	Say: 'Put your first model next to your final one. The difference is what you learned.' Give the Final Explanation checklist: ratio idea, tan chosen and why, eye height, consistency check, rounding error, real-world use. Circulate, and ask: 'What does your model show that your first one did not?' 'Where does your equation appear in the drawing?' WAIT TIME 10-15 seconds. Cold-call 4 learners to read their Final Explanation aloud, and ask the class to listen for the checklist items. Ask 2 learners to compare Lesson 1 and Final models aloud. Close with: 'Wanjiru and Otieno stood in different places and saw different angles, yet the tank is one tank. Why is trigonometry important?' Take 3	Model comparison (Lesson 1 versus Final) and Final Explanation. Placing the early and final models side by side makes conceptual growth visible, and self-explanation consolidates learning. Writing against a checklist ensures that the explanation integrates side naming, ratios, calculation, error and real-world relevance.	Score the Final Model and Explanation against the checklist: diagram labelled correctly; equation with 1.5 m; both learners' calculations leading to 7.5 m; ratio depends only on angle; rounding error explained; real-world application named. Success looks like a learner independently explaining why two different angles give one height. Exceeds: explains that a 1° reading error causes about 0.1 to 0.2 m difference at these distances. Approaching: leaves out eye height or cannot justify tan. Record these for follow-up and revision.

	angled triangle to find any unknown height or distance in our environment, without ever measuring it directly?' and 'Why is trigonometry important?' They must include: the ratio idea, the choice of tan, the eye height, the consistency check, the rounding error, and at least one real-world use (surveyors in the Kenyan highlands, engineers on the Thika Superhighway, or foresters in the Mau Forest). Finally, learners place Lesson 1 and Final Models side by side and write three differences, then share with a partner.	responses, and affirm that the ratio $\text{opposite} \div \text{adjacent}$ depends only on the angle, so a measured angle and a measured distance on the ground can replace a climb. Encourage learners to be proud: 'You have done what a surveyor does.'		
--	---	---	--	--

D. TEACHER REFLECTION

After the lesson, consider: (1) Could every learner draw a correctly labelled diagram, with the angle at the eye and the 1.5 m eye height shown separately, before calculating? Which learners still add or omit 1.5 m? (2) Did learners explain with confidence why Wanjiru (8 m, 37°) and Otieno (10 m, 31°) agree, using the idea that $\text{opposite} \div \text{adjacent}$ depends only on the angle? (3) How well did learners handle the 0.1 m discrepancy between Group Amani's 9.1 m and the tape's 9.0 m, and the 7.53 m and 7.51 m against 7.5 m? Did they see rounding to the nearest degree, rather than a flawed method, as the cause? (4) Which DQB questions were moved with strong evidence, and which with only assertion? (5) Did the Final Explanations name a real-world use in Kenya (surveyors, Thika Superhighway engineers, Mau Forest foresters)? (6) Which learners need extra support with calculator DEG mode or ratio choice, and what practice would help? Note questions left on the DQB (finding angles from sides, non-right-angled triangles) as pointers for future lessons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe? Wanjiru (8 m from the tower, 37°) and Otieno (10 m, 31°) got different angles. Both calculated the same height: 7.53 m and 7.51 m, which round to 7.5 m. Mr. Kamau's tape measured 7.5 m. Group Amani's flagpole calculation gave 9.1 m against the tape's 9.0 m.
What did I learn?	What did we learn? For a given angle, the ratio $\text{opposite} \div \text{adjacent}$ (tan) stays the same whatever the size of the triangle. A closer observer sees a larger angle. The height is found using $\text{height} = (\text{distance} \times \tan \text{angle}) + 1.5 \text{ m eye height}$. Angles read to the nearest degree cause small errors of about 0.1 m. A complete solution has a labelled diagram, the chosen ratio with a reason, the calculation, the answer with units and an explanation.
How does this explain the phenomenon?	How can we explain the phenomenon? The tank height can be found without climbing because the tower, the ground and the line of sight form a right-angled triangle. We measure the distance along level ground and the angle with a clinometer, and tan of the angle links them to the height above eye level. We add 1.5 m eye height to get 7.5 m. Both learners agree because the ratio depends only on the angle. Trigonometry is important because it lets surveyors, engineers and foresters find heights and distances they cannot measure directly.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.